
MATH BOOT CAMP NOTES

FOR INCOMING SSE MSc ECON STUDENTS

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Purpose of these notes

These notes are written for incoming SSE MSc Economics students as an introduction to some mathematical concepts and writing formal proofs during the summer. Before arriving at SSE, you are expected to have carefully read through these notes and worked with all the exercises, but not necessarily *master* this material.

The notes are mainly written as preparation for the course Math 5301, taught by Associate Professor Mark Voorneveld, but will also be useful for your other Term 1 courses. They are heavily inspired by excellent notes provided to me by my predecessors Johan Orrenius and Emil Bustos, along with Mark's course notes. Any remaining errors are my own.

Chapter 1 gives you an introduction to the course. Chapter 2 presents what constitutes a proof. Chapters 3–4 cover what constitutes a set and the canonical sets of numbers. While mastering formal proofs will take time, the underlying concepts should already be familiar to you. Chapters 5–12 introduce more advanced topics that you are not expected to master yet, but you should start to get a feel for.

Read the notes, try to make sense of the concepts, and attempt to solve the exercises. If you get confused, consult your favourite LLM for further help! None of the proofs in the suggested solutions are intended for memorisation – they are given to help you practice and understand how complete and rigorous arguments are built.

1 Introduction

Many incoming graduate students dread their first math course, expecting endless algebra or memorising formulas. But Math 5301 is different: it's about constructing clear, rigorous arguments from given definitions and results, just like in economic theory. The course mirrors foundations taught at top US PhD programs and teaches:

- **Mathematical Language:** To read and write advanced economic theory/models.
- **Logic of Economic Theory:** To see how definitions, theorems, and lemmas are built into arguments (of logical deduction) in a proof.
- **Mathematical Concepts:** To understand *why* results that underpin economic theory hold — not just *that* they hold — and how to prove them yourself.

Why do I need to know this?

If you lack interest in theory work or a PhD, you may wonder why this matters in an era dominated by AI and data. Even if you never write a formal proof again, clear thinking remains essential. Mathematical training is like lifting weights: the point of lifting weights at the gym is not to get good at weight-lifting, but to build strength for other parts of life. Likewise, math sharpens your ability to think, define problems, and follow arguments—whether in academic theory or real-world decisions.

What is an economic model?

Understanding economic models is one of the main reasons we study mathematical reasoning in this course. Purpose of economic models can roughly be divided as:

- **Pure theory models** — Built without data, focusing on internal logic and derivations from assumptions. They are the backbone of economic theory.
- **Structural models** — Used to estimate specific parameters in a fully specified economic model, often to test theories or simulate policy. Common in macro, IO, and the research-frontier of applied micro.
- **Stylized models** — Along with causal estimates, applied economists often include a *theoretical framework*. These are simple setups designed to illustrate mechanisms, not used directly in the empirical specification.
- **Econometric models** — Empirical models built to estimate relationships between economic variables using data. Although most econometrics take place in *probability spaces* with random variables, which is beyond the scope of this course, many mathematical concepts covered here can be seen as prerequisites for this.

On mastering mathematics

Solutions to the exercises are provided at the end of these notes, but I **strongly** encourage you to try each problem yourself first. Learning to write proofs is like learning to hammer a nail. A physics professor might understand the forces, angles, and materials involved—but still struggle to hit a nail straight. A carpenter, by contrast, becomes skilled through practice: missing, adjusting, and trying again until it becomes second nature. Math works the same way. It's not just a science—it's an *art*. You learn by doing: struggling, failing, and improving. Reading suggested solutions can help *after* you've tried, but passive reading alone won't build skill.

2 Proofs

2.1 Deduction

Imagine sitting by a pond, trying to predict the colour of the next swan to swim by.

Premise: All observed swans have been white.

Conclusion: The next swan will be white.

This is an example of **induction** — a pattern-based guess, not a logical guarantee.

Example 2.1 (Induction). ***Premise:** In a large number of empirical settings, raising the marginal labour-income tax rate has caused reductions in hours worked. **Conclusion:** Raising the labour-income tax rate reduces labour supply in any setting.*

Now instead, consider the logical step:

Premise: All swans are white.

Conclusion: The next swan will be white.

This is **deduction** — if the premise is true, the conclusion must follow. In this course, we are concerned with deduction — understanding how to derive logically valid conclusions from clearly stated assumptions.

Example 2.2 (Deduction). ***Premise:** A rational agent chooses leisure $\ell \in [0, T]$ to maximise quasi-linear utility $u(c, \ell) = c + v(\ell)$, with v strictly concave and increasing, subject to $c = w(1 - \tau)(T - \ell)$, where $w > 0$ is the wage and $\tau \in [0, 1]$ is the tax rate. **Conclusion:** Under above assumptions, optimal labour supply $h^*(\tau) = T - \ell^*(\tau)$ is strictly decreasing in τ , i.e. raising the labour-income tax rate reduces labour supply.*

2.2 Logical Connectives

One can build statements from simpler parts using **logical connectives**. In particular, let p and q be two arbitrary *propositions* (statements), regarding some real number x , that can be either true or false. Define p as the proposition “ $x > 0$ ”.

- *Negation* ($\neg p$): This means “not p ”. The proposition $\neg p$ is true exactly when p is false. **Example:** $\neg p$ is the proposition “ $x \leq 0$ ”.
- *Conjunction* ($p \wedge q$): This means “ p and q are both true”. The conjunction is true only when *both* p and q are true. **Example:** Define q as “ $x < 1$ ”. Then $p \wedge q$ is the proposition “ $0 < x < 1$ ”.

- **Disjunction** ($p \vee q$): This means “ p is true or q is true (or both)”. It is false only when *both* p and q are false. **Example:** Define q as “ $x < -1$ ”. Then the proposition $p \vee q$ means “ x is not in the interval $[-1, 0]$ ”.

Unless explicitly stated otherwise, mathematicians use an *inclusive* “or”. The statement “ p or q is true” means that *at least* one of p and q is true, not that *exactly* one is true!

In this course, you will often see statements regarding *implications*: “if p then q ”, denoted $p \Rightarrow q$. This is declared to be false when p is true, but q is false, and is declared to be true in all other cases. For example, if p is the proposition “ $x > 0$ ” and q is “ $x^2 > 0$ ”, then the implication $p \Rightarrow q$ is true. Note that this does not say anything about q if p is not true, or whether p is true or not.

In addition, you will encounter *equivalence* statements ($p \Leftrightarrow q$) like “ p if and only if q ”. An equivalence statement is true when p and q have the same truth value (both true or both false). It can also be described as $(p \Rightarrow q) \wedge (q \Rightarrow p)$ being true. For example, $x^2 = 0 \Leftrightarrow x = 0$.

Finally, throughout this course, we will take as given for proofs by *contrapositive*:

Proposition 2.1.

$$p \Rightarrow q \quad \Leftrightarrow \quad \neg q \Rightarrow \neg p$$

Consider the previous example of $p : x > 0$ and $q : x^2 > 0$, so $p \Rightarrow q$. Then it follows that $\neg q \Rightarrow \neg p$, i.e. $x^2 \leq 0 \Rightarrow x \leq 0$. However, note that $x \leq 0 \not\Rightarrow x^2 \leq 0$.

Example 2.3 (Necessity and Sufficiency). Consider the statement “If it rains, then the street gets wet”. Let R denote the proposition “it is raining” and W the proposition “the street is wet”. Then the implication can be written as:

$$R \Rightarrow W.$$

That is, rain is a **sufficient** condition for the street to be wet. Using contrapositive logic, we can also write:

$$\neg W \Rightarrow \neg R.$$

In words: if the street is not wet, then it did not rain. However, rain is not a **necessary** condition for the street being wet: it is **not** the case that $\neg R \Rightarrow \neg W$. In words: the street can be wet even if it didn’t rain (e.g. a sprinkler was on).

2.3 How to prove something

Logical connectives guide how we combine assumptions and conclusions in proofs:

Direct Proof: To *prove* an implication ($p \Rightarrow q$), we assume p and show that q follows. To *disprove* an implication, we find a case where p is true but q is false, i.e. we find a counter-example. To prove an equivalence ($p \Leftrightarrow q$), we prove both $p \Rightarrow q$ and $q \Rightarrow p$.

As you will see throughout this course, proving that a statement holds is a lot more difficult than proving that a statement does not hold. To disprove a statement, you need to find a single counter-example where it does not hold. To prove a statement, you need to show that there does not exist a single counter-example. You can see this asymmetry clearly in **Exercise 9.1**.

Proof by Contradiction: If we want to show that p is true, it is the same as showing $\neg p$ is false. We assume p is false and then show that we reach a contradiction somewhere and then deduce that p must be true. For a classic example, see **Exercise 4.1**.

Proof by Induction: Assume we have a sequence of statements $p(n)$ with $n \in \mathbb{N}$. Then, if we show that (i) $p(1)$ is true, and (ii) $p(n+1)$ is true whenever $p(n)$ is true, then we have shown that $p(n)$ is true for all n .

Exercise 2.1. Show that for all integers $n \geq 1$,

$$2 + 4 + 6 + \cdots + 2n = n(n+1)$$

2.4 What to Prove?

Proving a proposition is much like making a dish, such as lasagna. If you had to run back and forth to the store each time you needed the next ingredient, the cooking would be chaotic and exhausting. Instead, you first gather all the necessary ingredients that are likely to be useful. Only then do you start assembling the dish, carefully layering the pieces together.

In our examples above, we have not necessarily gone into issues like “what exactly is a *number*?”, “what is a *swan*?”, “what is the colour *white*?”. To make formal proofs, all such issues of definition must be resolved before starting to prove e.g. an implication.

The rest of these notes will familiarise you with mathematical definitions and theorems (the ingredients), and you will use the exercises to practice how to use clearly defined propositions and premises to draw conclusions (making the dish). Good luck!

3 Sets

In this section, we learn *naive set theory*, based on *Appendix A* in Mark's notes.

3.1 Collection of elements

Sets are (unordered) collections of *elements*, for example, $\{1, 2, 3\}$ or $\{a, b\}$. The elements can be all sorts of things, such as numbers, functions, names, or other sets. When we describe sets, we either list the elements directly, such as in the examples above, or define a condition for an element to be part of a set (see below).

3.2 Set Operations

Now that we have defined sets, we can start discussing relations between sets. To say that element x belongs to the set X , we write $x \in X$. If an element x is not in X , we write $x \notin X$. We denote “for all elements in set X ” as $\forall x \in X$ and “there exists some element in X ” as $\exists x \in X$.

Definition 3.1 (Subset). *If every element in A also belongs to B , then A is a **subset** of B . Formally, if $x \in A \Rightarrow x \in B$, then $A \subseteq B$.*

Definition 3.2 (Proper subset). *If $A \subseteq B$ but $B \not\subseteq A$, then A is a **proper subset** of B . Formally, if $x \in A \Rightarrow x \in B$ and $\exists y \in B$ such that $y \notin A$, then $A \subset B$.*

Definition 3.3 (Identical sets). *If $A \subseteq B$ and $B \subseteq A$, then A and B are **identical**. Formally, $A = B$ iff $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$. Here, “iff” refers to the equivalence statement “if and only if”.*

Throughout these notes we write $:=$ when we *define* a new object, and $=$ when the equality is something we have derived or are asked to verify.

Example 3.1. *Denote the set of all students at SSE as X . Denote the set of students in Year 1 MSc Economics as E , defined as $E := \{x \in X : x \text{ is Year 1 MSc Econ}\} \subset X$ meaning “ E is a set of every SSE student **such that** the student is a Year 1 MSc Econ student”.*

The set of elements that are in A or B (or both!) is called the *union* of A and B , written as $A \cup B$. Likewise, if we have n sets $A_1, \dots, A_n$, their union $A_1 \cup \dots \cup A_n$ consists of all elements that belong to *any* of these sets. This may be written $\cup_{i=1}^n A_i$. Equivalently, we may define the set $I := \{1, \dots, n\}$ and write the union as $\cup_{i \in I} A_i$.

Example 3.2. *Denote the set of students at different programs at SSE as $D_1, \dots, D_n$. The set of students at SSE is the union of all of these: $X = \cup_{i=1}^n D_i$.*

A set of real numbers between two endpoints is an *interval*: we write $(a, b) := \{x : a < x < b\}$ and $[a, b] := \{x : a \leq x \leq b\}$, and likewise for the mixed cases $[a, b)$ and $(a, b]$ – a square bracket includes the endpoint, a round one does not. An unbounded side takes a round bracket and the symbol ∞ , as in $[0, \infty)$.

Example 3.3. Denote some set of real numbers on the real interval as $D_i := (-i, i)$, so e.g. $D_1 = (-1, 1)$. The union of all such intervals is $\cup_{i \in \mathbb{N}} D_i = (-\infty, \infty)$.

The set of elements that are in A and in B is called the *intersection* of A and B , written as $A \cap B$. Finally, we denote the empty set $\{\}$ as $\emptyset$ and it is a subset of every set.

Example 3.4. No students attend all programs at SSE. So $\cap_{i=1}^n D_i = \emptyset$, i.e. the set of all students at SSE that attends every program is the empty set $\emptyset := \{\}$.

Example 3.5. The intersection of all intervals $D_i := (-i, i)$ is $\cap_{i=1}^{\infty} D_i = (-1, 1)$.

The set of elements in X that are not in A is written $X \setminus A$. If $A \subset X$, then we say that the *complement* of A is $X \setminus A$, denoted A^c . Importantly, note that the complement of A , denoted A^c , is defined in relation to the “larger” set X .

Example 3.6. SSE students that are not Year 1 MSc Econ students can be written as E^c .

3.3 The set of all subsets

So far, the elements of our sets have been points – numbers, students, bundles. But elements can themselves be sets. The most important instance is the collection of *all* subsets of a given set (where each element is a subset).

Definition 3.4 (Power set). The **power set** of a set X , denoted $\mathcal{P}(X)$, is the set whose elements are all subsets of X :

$$\mathcal{P}(X) := \{A : A \subseteq X\}$$

Two points of care. First, “ $\subseteq$ ” inside $\mathcal{P}(X)$ becomes “ $\in$ ” outside it: $A \subseteq X$ if, and only if, $A \in \mathcal{P}(X)$. Second, $\emptyset \in \mathcal{P}(X)$ and $X \in \mathcal{P}(X)$ always.

Example 3.7. Let $X = \{a, b\}$. Then

$$\mathcal{P}(X) = \{ \emptyset, \{a\}, \{b\}, \{a, b\} \}$$

which has $2^2 = 4$ elements. In general, if X has n elements, then $\mathcal{P}(X)$ has 2^n elements – which is where the alternative notation 2^X comes from. Note that $a \in X$ but $a \notin \mathcal{P}(X)$: the elements of $\mathcal{P}(X)$ are sets, not points. What lives in $\mathcal{P}(X)$ is the singleton $\{a\}$.

4 Numbers

We are used to working with numbers to count and label things. For example, a set with two elements is said to have a *cardinality* of two. These basic counting numbers $1, 2, 3, \dots$ are called the *natural numbers* $\mathbb{N}$:

$$\mathbb{N} := \{1, 2, 3, \dots\} \quad (\text{Natural numbers})$$

First, note that in this course, we will use the definition of the natural numbers that does not include zero. Second, sets are not generally “equipped” with an *order* “ $>$ ”, but we will here assume sets of numbers “inherit” the natural order we are used to, e.g., $3 > 2$.

Next, we extend the natural numbers to include zero and the negatives of natural numbers. This gives the set of *integers* $\mathbb{Z}$. Formally, we define

$$\mathbb{Z} := \{0\} \cup \mathbb{N} \cup \{-n : n \in \mathbb{N}\} \quad (\text{Integers})$$

The above might look tricky at first, but can be read as “the set of integers $\mathbb{Z}$ is the union of three sets: a singleton set with a zero, the set of natural numbers, and a set consisting of the negative of each element in the natural numbers (negative numbers)”. We can extend the integers by allowing *ratios* of integers: the *rational numbers*.

Definition 4.1. Denote the set of rational numbers as $\mathbb{Q}$. For any $x \in \mathbb{Q}$, we can write $x = \frac{m}{n}$ for some $m, n \in \mathbb{Z}$ with $n \neq 0$. Formally,

$$\mathbb{Q} := \left\{ \frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{Z} \setminus \{0\} \right\} \quad (\text{Rational numbers})$$

Exercise 4.1. Formally, x is even if, and only if, $x = 2k$ for some $k \in \mathbb{Z}$. First, show that $x \in \mathbb{Z}$ is even if, and only if, x^2 is even. Then, show that $\sqrt{2}$ is not rational.

Now we get to the workhorse numbers of this course, the *real numbers*. Geometrically, the rationals sit as a dense scatter of points on the number line – but with countless gaps between them, at places like $\sqrt{2}$ and π . The real numbers fill those gaps in, and a gap can be named either by where it splits the rationals, or by a run of rationals crowding ever nearer to it, such as $1, 1.4, 1.41, 1.414, \dots$ for $\sqrt{2}$. The second is just a decimal expansion, terminating like $\frac{3}{8} = 0.375$ or endless like $\pi = 3.141592\dots$. We denote the reals by $\mathbb{R}$ and the non-negative reals as $\mathbb{R}_+ := [0, \infty)$. For a rigorous treatment, see: [Wikipedia: Construction of the real numbers](#).

Thus, we may summarize the construction as

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

5 Spaces

The next several chapters introduce different kinds of *spaces*. To help you keep track, here is a preview of how they relate. The key point is that each kind of space is just a set equipped with some *operators*; sometimes one kind of structure automatically gives you another for free, and sometimes the kinds of spaces are independent of each other.

Take a set X , simply a collection of elements. Then:

- **Vector space:** A (real) vector space V is a set X equipped with addition $(+)$ and real scalar multiplication $(\cdot)$ such that V is *closed* under both, i.e. vectors can be added together and rescaled without leaving the space. We can write a vector space as a *triple* $V := (X, +, \cdot)$.
- **Inner product space:** An inner product space I is a vector space V equipped with an inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$. Notationally, read the previous as “an inner product is a *mapping* that takes two elements from V and spits out a number”. In lower dimensions, the intuition is that an inner product $\langle x, y \rangle$ gives us the *angle* of two vectors x, y . We can write this inner product space as $I := (V, \langle \cdot, \cdot \rangle)$.
- **Normed vector space:** A normed vector space N is a vector space V equipped with a norm $\| \cdot \| : V \rightarrow \mathbb{R}_+$, where $\|v\|$ measures the length of a vector v . We can write it as $N := (V, \| \cdot \|)$. We can always derive a norm from an inner product by setting $\|v\| := \sqrt{\langle v, v \rangle}$. The reverse is not always possible.
- **Metric space:** A metric space M is a set X equipped with a distance function $d : X \times X \rightarrow \mathbb{R}_+$, where the distance function $d(x, y)$ measures the distance between elements x, y . We can write this as $M := (X, d)$. Note that, on its own, a metric space is structurally the simplest of the four – it requires no algebra, just a set and a way to measure distances. But whenever we already have a normed vector space in hand, we can derive a metric for free by setting $d(x, y) := \|x - y\|$. The reverse is not always possible.

Schematically, the derivations only go one way:

$$\text{vector space} \xrightarrow{\text{add } \langle \cdot, \cdot \rangle} \text{inner product space} \xrightarrow{\|v\| := \sqrt{\langle v, v \rangle}} \text{normed vector space} \xrightarrow{d(x, y) := \|x - y\|} \text{metric space}$$

In each chapter ahead, we will pin down one of these structures. Once we have metric spaces in hand, we will then study *mappings* between metric spaces.

A last way of building a set from two others is to pair up their elements. The **Cartesian product** of X and Y is the set of ordered pairs $X \times Y := \{(x, y) : x \in X, y \in Y\}$, and we write $X^2 := X \times X$, so that the plane $\mathbb{R}^2$ is exactly $\mathbb{R} \times \mathbb{R}$. More generally, $\mathbb{R}^n$ is the set of ordered lists $(x_1, \dots, x_n)$ of n real numbers.

Example 5.1 (One set, four kinds of space). *To make the relationships above concrete, take the simplest non-trivial set: $X = \mathbb{R}^2$. Watch how each kind of structure can be put on X .*

- **As a set.** $X = \mathbb{R}^2 = \{(x_1, x_2) : x_1, x_2 \in \mathbb{R}\}$. No operations yet — just elements.
- **As a vector space.** Equip X with coordinate-wise addition and scalar multiplication: $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$, $\alpha(x_1, x_2) = (\alpha x_1, \alpha x_2)$. Both operations land back in $\mathbb{R}^2$, so X is closed under both. We can now add and rescale elements.
- **As an inner product space.** Equip X with the dot product $\langle x, y \rangle = x_1 y_1 + x_2 y_2$.
- **As a normed vector space.** The norm derived from this inner product is $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x_1^2 + x_2^2}$ — the familiar Euclidean length (from origo).
- **As a metric space.** The metric derived from this norm is the Euclidean distance $d(x, y) = \|x - y\| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$. We can now measure the distance between two elements.

Same set $\mathbb{R}^2$ throughout, just different structures put on it. The same set $\mathbb{R}^2$ could also be turned into a different metric space directly, with no algebra involved at all — for example by using the discrete metric $d_0(x, y) = 1$ whenever $x \neq y$ and 0 otherwise. The standard choice (that we are used to working with) for $\mathbb{R}^2$ just happens to be the Euclidean one above.

Before we dig in, let me introduce someone you will see throughout these notes.

ECON APPLICATION: Meet Anna

Anna is a consumer choosing between two goods. Her *consumption bundle* is an ordered pair $x = (x_1, x_2) \in \mathbb{R}^2$, where x_1 is the quantity of good 1 and x_2 is the quantity of good 2 — exactly the kind of vector this chapter is about. The unit prices are $p = (p_1, p_2) = (1, 2)$, her income is $y = 10$, and her preferences are represented by a utility function $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ on the non-negative quadrant $\mathbb{R}_+^2 := \{x \in \mathbb{R}^2 : x_1, x_2 \geq 0\}$, such that

$$u(x_1, x_2) = \sqrt{x_1 x_2}.$$

Eventually we want to ask: which bundle $x^* = (x_1^*, x_2^*)$ maximises Anna's utility subject to her budget? We cannot answer that yet — we don't even have the language. But each chapter from here on will give us one tool that her problem needs, and by the end of these notes we'll watch her solve it.

6 Vectors

When getting familiar with vector spaces, starting with the most simple notion of a vector is useful. I find this course [Essence of Linear Algebra](#) illuminating to get an intuitive sense of vector spaces and matrices. Begin with: [3Blue1Brown: Vectors](#).

6.1 Ordered List Vectors

A *set* is an unordered collection of elements: for example, $\{a, b\} = \{b, a\}$. But sometimes order matters – for instance, when working with coordinates. In such cases, we use *ordered pairs*, such as (a, b) , where the order of elements is essential.

Vectors can be visualised as arrows in the plane. Define a collection of vectors $E := \{e_1, e_2\}$, where $e_1 := \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$ and $e_2 := \begin{bmatrix} 0 & 1 \end{bmatrix}^\top$. The notation $\top$ is called a transpose, see **Section 7.4**. Probably, this is the *basis* you use when you interpret the below vector $\mathbf{v} = (2, 1)$ – a *linear combination* of e_1, e_2 , each scaled by a number and then added together – as having *coordinates*, see **Figure 6.1**. It is called the *standard* or *orthonormal* basis: the two directions are perpendicular with length one. These notes use it throughout.

$$\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (6.1)$$

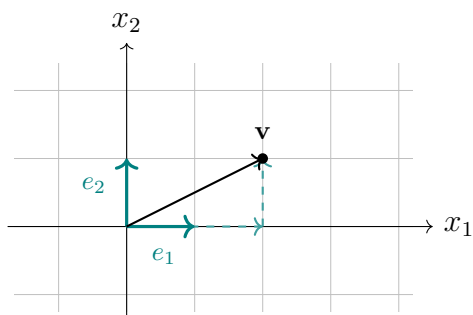
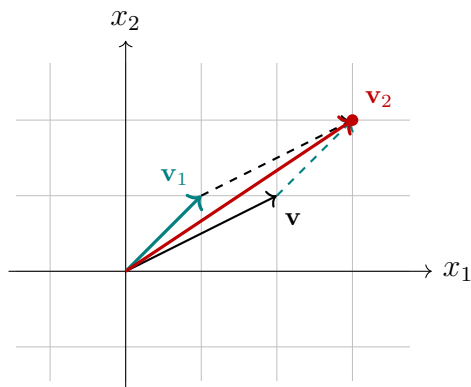
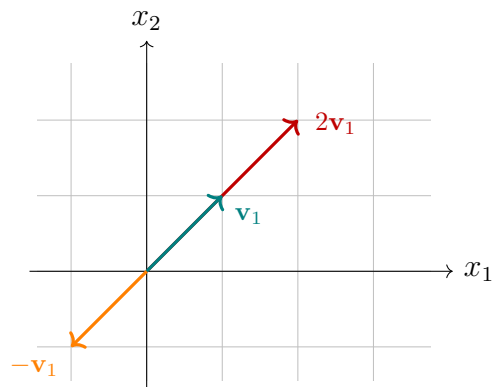


Figure 6.1: E as a basis: $\mathbf{v}$ is two steps east and one step north.

Example 6.1 (Adding and scaling). Vectors can be added together. Adding the vector $\mathbf{v}_1 = (1, 1)$ to our vector $\mathbf{v}$ gives $\mathbf{v}_2 = \mathbf{v}_1 + \mathbf{v} = (3, 2)$, and the order does not matter: laying a copy of $\mathbf{v}_1$ on the tip of $\mathbf{v}$ takes us to the same place as laying a copy of $\mathbf{v}$ on the tip of $\mathbf{v}_1$, see **Figure 6.2a**. Scaling by a positive number makes a vector longer or shorter while keeping its direction; a negative scalar flips its direction. For example, $2\mathbf{v}_1 = (2, 2)$ and $-\mathbf{v}_1 = (-1, -1)$, see **Figure 6.2b**.



(a) $\mathbf{v}_2 = \mathbf{v}_1 + \mathbf{v} = (3, 2)$



(b) Scaling $\mathbf{v}_1$ by 2 and by -1

Given a collection of vectors, we can ask how much of the space we can reach using addition and scaling:

Definition 6.1 (Linear combination and span). Let $v_1, \dots, v_k \in \mathbb{R}^n$. A **linear combination** of them is any vector $\alpha_1 v_1 + \dots + \alpha_k v_k$ with $\alpha_1, \dots, \alpha_k \in \mathbb{R}$. Their **span** is the set of all such combinations: $\text{span}\{v_1, \dots, v_k\} := \{\alpha_1 v_1 + \dots + \alpha_k v_k : \alpha_1, \dots, \alpha_k \in \mathbb{R}\}$

Definition 6.2 (Linear independence). The vectors $v_1, \dots, v_k$ are **linearly independent** if $\alpha_1 v_1 + \dots + \alpha_k v_k = 0$ only when $\alpha_1 = \dots = \alpha_k = 0$. Otherwise they are **linearly dependent**, and at least one vector can be written as a linear combination of the others.

The two ideas together are what make a *coordinate list* unambiguous. Here e_1 and e_2 are linearly independent – neither is a multiple of the other – and $\text{span}\{e_1, e_2\} = \mathbb{R}^2$, so every arrow in the plane can be written as $\alpha_1 e_1 + \alpha_2 e_2$, and in exactly one way.

Neither test is automatic, though, and a collection can fail both at once.

Example 6.2 (A collection that fails both tests). Take $u_1 := (1, 0, 0)$, $u_2 := (0, 1, 0)$ and $u_3 := (1, 1, 0)$ in $\mathbb{R}^3$. They are not linearly independent, since $u_1 + u_2 - u_3 = 0$ with coefficients that are not all zero. The third vector is therefore redundant: anything we can reach with it, we could already reach without it. Nor do they span $\mathbb{R}^3$, because every linear combination of them has a third coordinate of 0,

$$\alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 = (\alpha_1 + \alpha_3, \alpha_2 + \alpha_3, 0),$$

so their span is the flat x_1 - x_2 plane of **Figure 6.3** rather than the whole space. The vector $w := (0, 0, 1)$ is an ordinary vector of $\mathbb{R}^3$ that no combination of the three can reach.

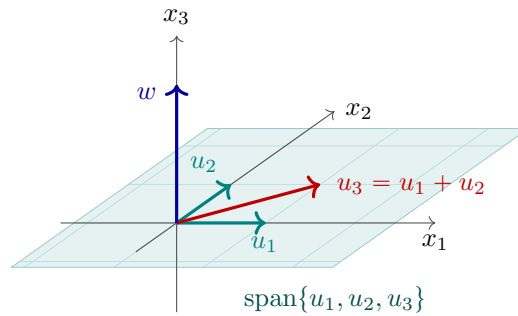


Figure 6.3: Three vectors in $\mathbb{R}^3$ and their span

6.2 Vector spaces

A *vector space* is a set V whose elements (“vectors”) you can add together and multiply with numbers (“scalars”) to produce new vectors in V . So far, we’ve discussed vectors as ordered lists of numbers. But a vector can also be a number, a matrix, or a function. What matters is that we can construct a well-functioning space of them.

Definition 6.3 (Vector spaces). A (real) **vector space** V is a set (of “vectors”) on which two operations, addition $+$ and scalar multiplication $\cdot$, are defined such that

- V is closed under addition: for each pair of elements $x, y \in V$ there is a unique element $x + y$, the sum of x and y , in V ,
- V is closed under scalar multiplication: for each $x \in V$ and each $\alpha \in \mathbb{R}$, there is a unique element αx , the (scalar) product of α and x , in V .

Moreover, addition and scalar multiplication satisfy standard arithmetic properties.

Exactly what is meant by “standard arithmetic properties” is governed by the eight “vector axioms”, which will be more closely defined later on in the course. Below are some simple examples of sets that can be turned into vector spaces using our basic notions of addition and scalar multiplication (and some that can’t!).

Example 6.3 (The Real Line). The set $\mathbb{R}$ equipped with the standard addition and scalar multiplication is a real vector space. Vectors are real numbers and the zero element is 0. Addition is standard addition, and scalar multiplication is just multiplying by a real number: for $x, y, \alpha \in \mathbb{R}$, we have $x + y \in \mathbb{R}$ and $\alpha x \in \mathbb{R}$.

Example 6.4 (The Interval $(0,1)$). Consider the set $(0,1)$ equipped with the same $+$ and $\cdot$. This is **not** a real vector space. It is not closed under addition: $0.8 + 0.5 = 1.3 \notin (0,1)$. Nor

is it closed under scalar multiplication: $2 \cdot 0.6 = 1.2 \notin (0, 1)$.

Example 6.5 (The Rational Numbers $\mathbb{Q}$). The set $\mathbb{Q}$ of rational numbers equipped with the standard $+$ and $\cdot$ is not a real vector space. To be a real vector space, it must be closed under real scalar multiplication – including scaling with irrational numbers. But $\sqrt{2} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}} \notin \mathbb{Q}$. Since $\mathbb{Q}$ is not closed under scalar multiplication, it is not a real vector space.

Example 6.6 ($\mathbb{R}^2$). The set $\mathbb{R}^2$ consists of ordered pairs (x_1, x_2) of real numbers. The zero element is $(0, 0)$. Addition and scalar multiplication are defined coordinatewise: for two ordered pairs $(x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$, we have $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ and $\alpha(x_1, x_2) = (\alpha x_1, \alpha x_2)$. For example, $(1, 3) + (2, 5) = (3, 8)$ and $2 \cdot (1, 4) = (2, 8)$.

Example 6.7 (Quadratic Polynomials). The set $P_2(\mathbb{R})$ of real polynomials of degree at most 2 is a vector space. The zero element is the zero polynomial $p(x) = 0$. Each element is a function of the form $p(x) = a_0 + a_1x + a_2x^2$ with $a_0, a_1, a_2 \in \mathbb{R}$. Addition and scalar multiplication are defined coefficient-wise. For two functions $p(x), q(x) \in P_2(\mathbb{R})$, where $q(x) = b_0 + b_1x + b_2x^2$ and $\alpha \in \mathbb{R}$, we have $(p+q)(x) = (a_0+b_0) + (a_1+b_1)x + (a_2+b_2)x^2$ and $(\alpha p)(x) = \alpha a_0 + \alpha a_1x + \alpha a_2x^2$. For example, if $q_1(x) := 2x$ and $q_2(x) := x^2 + x + 1$, then $p(x) = x^2 + 3x + 1$, so $p(x) \in P_2(\mathbb{R})$, see **Figure 6.4**.

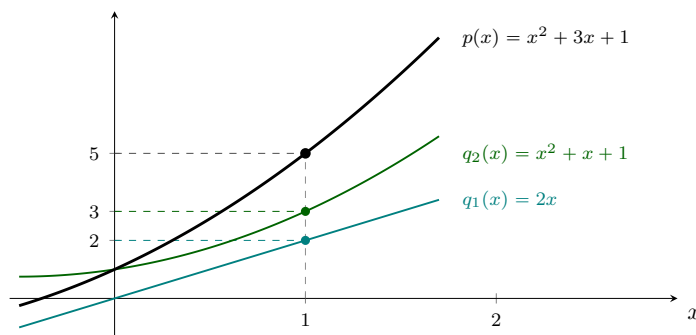


Figure 6.4: Adding two polynomials adds their graphs pointwise: at $x = 1$, $p(1) = q_1(1) + q_2(1) = 2 + 3 = 5$

6.3 Subspaces

We often look at subsets of a vector space V with additional nice properties. If such a smaller set – with addition and scalar multiplication as in the larger set V – satisfies all properties of a vector space, it is called a (linear) *subspace*. Using the theorem below, one can verify that a given subset W is a subspace.

Definition 6.4 (Subspaces). A subset W of vector space V is a (linear) **subspace** of V if W itself (inheriting $+$ and $\cdot$ from V) is a vector space.

Theorem 6.1. A subset W of a vector space V is a subspace if and only if it satisfies the following three properties:

- i) W contains the zero vector from V : $0 \in W$,
- ii) W is closed under addition: $x + y \in W$ whenever $x \in W$ and $y \in W$,
- iii) W is closed under scalar multiplication: $\alpha x \in W$ whenever $x \in W$ and $\alpha \in \mathbb{R}$.

Exercise 6.1. Use the theorem above to show that

- a) The empty set $W_0 := \emptyset$ is not a subspace of $\mathbb{R}^2$.
- b) The set $W_1 := \{x \in \mathbb{R}^2 : x_1 = 0 \text{ or } x_2 = 0\}$ is not a subspace of $\mathbb{R}^2$.
- c) The set $W_2 := \{x \in \mathbb{R}^2 : x_1, x_2 \in \mathbb{N}\}$ is not a subspace of $\mathbb{R}^2$.
- d) The set $W_3 := \{x \in \mathbb{R}^2 : 3x_1 - 4x_2 = 0\}$ is a subspace of $\mathbb{R}^2$.
- e) The set $W_4 := \{f \in P_2(\mathbb{R}) : f(x) = a + bx \text{ for some } a, b \in \mathbb{R}\}$ is a subspace of the set $P_2(\mathbb{R})$ from **Example 6.7**.

6.4 Angles and lengths

So far, vectors are objects we can add and scale. To talk about *angles* between vectors, or to relate the length of a vector to its components, we need a third operation: an **inner product**. A vector space equipped with an inner product is called an *inner product space*.

Example 6.8 (The inner product on $\mathbb{R}^n$). The standard inner product on $\mathbb{R}^n$ is the **dot product**: for $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$,

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i.$$

Other common notation includes $x \cdot y$ and $x^\top y$ – the latter because the inner product on $\mathbb{R}^n$ is the matrix product of the row vector $x^\top$ with the column vector y . We will use this connection in the next chapter when discussing matrix multiplication. For a geometric interpretation, see [here](#).

ECON APPLICATION: Anna's expenditure

The cost of Anna's bundle $x = (x_1, x_2) = (3, 5)$ under prices $p = (1, 2)$ is the inner product $\langle p, x \rangle = p_1x_1 + p_2x_2 = 3 + 2 \cdot 5 = 13$. More generally, with n goods and unit prices $p = (p_1, \dots, p_n)$, the cost of a bundle $x = (x_1, \dots, x_n)$ is $\langle p, x \rangle = \sum_i p_i x_i$.

Just as the absolute value $|x|$ measures the size of a scalar, we can define the *length* (or *norm*) of a vector in $\mathbb{R}^n$ using the Euclidean norm, sometimes referred to as the ℓ^2 norm:

$$\|\mathbf{v}\| = \sqrt{\langle v, v \rangle} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

If we equip a vector space with a norm, it becomes a *normed vector space*.

Example 6.9. In $\mathbb{R}^2$ and $\mathbb{R}^3$, think of this as the Euclidean distance from the origin to the “point” of the vector. In $\mathbb{R}^2$, the vector $\mathbf{v}' = (3, 4)$ has length $\|\mathbf{v}'\| = \sqrt{3^2 + 4^2} = 5$. In $\mathbb{R}^3$, the length of $\mathbf{v}_3 = (3, 4, 2)$ is $\|\mathbf{v}_3\| = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{29}$. Its shadow on the x_1 - x_2 floor is exactly $\mathbf{v}'$, so the plane answer 5 is one leg and the height 2 the other, see **Figure 6.5**.

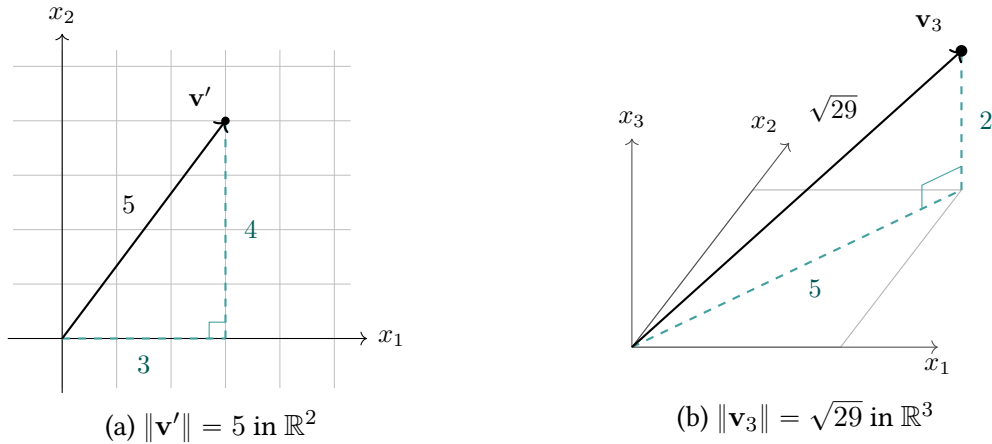


Figure 6.5: The norm as a diagonal: in $\mathbb{R}^3$ it is the plane answer 5 with one more leg

7 Matrices

One particular type of vector space is a space of *matrices*. A matrix is a collection of rows and columns, organised in a rectangular fashion. We denote matrices by capital letters, A , and their element in row i and column j as a_{ij} .

7.1 Addition and Scalar Multiplication

First, we note that a vector space of matrices is simply a set of matrices (vectors), equipped with addition and scalar multiplication that obeys standard arithmetic rules.

Example 7.1 (2×3 Matrices). *The set $\mathbb{R}^{2 \times 3}$ of 2×3 real matrices is a vector space. If*

$$M := \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \end{bmatrix}, \quad N := \begin{bmatrix} n_{11} & n_{12} & n_{13} \\ n_{21} & n_{22} & n_{23} \end{bmatrix},$$

then addition and scalar multiplication are defined entrywise:

$$(M + N)_{ij} = m_{ij} + n_{ij}, \quad (\alpha M)_{ij} = \alpha m_{ij}.$$

The zero element is the 2×3 matrix with all entries equal to 0.

The addition operator for matrices is a commutative operation, so $A + B = B + A$.

Example 7.2 (Addition and scalar multiplication of matrices). *We have:*

$$A := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad A + B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \quad 5 \cdot A = \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix}$$

7.2 Matrix multiplication

Recall the inner product on $\mathbb{R}^n$ from **Chapter 6**. *Matrix multiplication* extends this idea: each entry of a matrix product is itself an inner product of a row of the first matrix with a column of the second. As you will see below, matrix multiplication is not like number multiplication: it's not commutative, so we may have $A \cdot B \neq B \cdot A$. Furthermore, it only works when the “inner dimensions” match. If C is an $n \times m$ matrix and D is an $m \times p$ matrix, then the product $C \cdot D$ is defined and results in an $n \times p$ matrix. What matters is that the number of columns m of the first matrix must match the number of rows m of the second matrix.

Example 7.3 (Matrix multiplication). Consider the matrices

$$C := \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}_{2 \times 3}, D := \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}}_{3 \times 2} \implies C \cdot D = \begin{bmatrix} 1 \cdot 1 + 3 \cdot 1 & 2 \cdot 1 + 3 \cdot 1 \\ 4 \cdot 1 + 6 \cdot 1 & 5 \cdot 1 + 6 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 10 & 11 \end{bmatrix}$$

Each entry is one inner product: entry (i, j) of $C \cdot D$ pairs row i of C with column j of D , multiplying term by term and summing. The top-left entry pairs $(1, 2, 3)$ with $(1, 0, 1)$ and gives $1 \cdot 1 + 2 \cdot 0 + 3 \cdot 1 = 4$. This is why the inner dimensions have to match: a row and a column must have the same length. We take the number of rows from C and of columns from D and get a 2×2 matrix. If we instead calculate $D \cdot C$, we get a 3×3 matrix.

Example 7.4 (Multiplying a vector with a matrix). Consider the vector $v = (2, 1)$ and the matrix A from **Exercise 7.2**. We get:

$$Av = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 \\ 3 \cdot 2 + 4 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix} \quad (7.1)$$

7.3 Matrices as linear transformations

We now know how to *calculate* with a matrix. But what does a matrix actually *do*? As explained in [3Blue1Brown: Linear Transformations](#), multiplying by a matrix M is a **linear transformation** (a function), eating a vector and returning a vector, written $T(v) = Mv$. It is **linear** because it can be pictured as moving the whole plane at once, grid lines staying parallel and evenly spaced with origin fixed. Every vector can be written as $v = v_1e_1 + v_2e_2$ and $Mv = v_1Me_1 + v_2Me_2$, so knowing where e_1 and e_2 land tells us where any vector v lands. A matrix is the table of the “landing spots” for the basis vectors: they are its columns.

Example 7.5 (Multiplying a vector with a matrix). Take the *shear*

$$S := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad Se_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad Se_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

While e_1 stays put, e_2 slides one step east. So $v = (2, 1)$, which takes two of e_1 (which stays put) and one of e_2 (which slides one step east), will itself slide one step east. We get $Sv = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, as shown in **Figure 7.1**. If we instead considered a vector $w = (1, 3)$, we would get $Sw = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$.

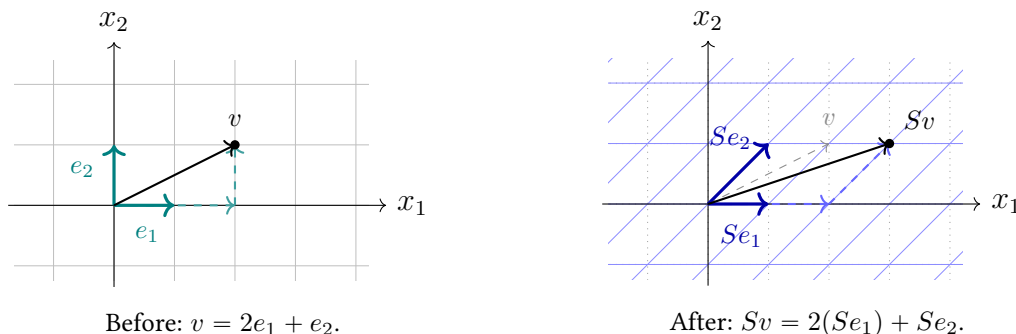


Figure 7.1: The shear S read as a motion of the whole plane

Example 7.6 (Matrix multiplication as composition). *What happens if we multiply a vector with a matrix that resulted from multiplying two matrices together? Take S from **Example 7.5** together with the **rotation** matrix $R := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ that turns a vector a quarter turn anticlockwise. You can see this rotation in its columns: e_1 lands north at $(0, 1)$ and e_2 west at $(-1, 0)$. Multiplying R with S gives:*

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}.$$

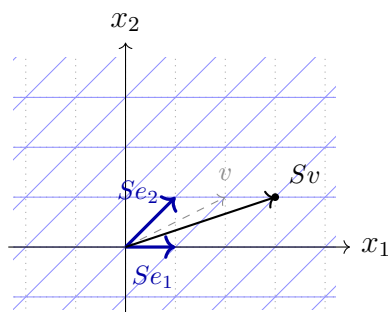
*Apply both to $v = (2, 1)$ in turn: $Sv = (3, 1)$, then $R \cdot (3, 1) = (-1, 3)$. The single matrix RS gets there in one step, since $(RS)v = 2\begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1\begin{pmatrix} -1 \\ 1 \end{pmatrix} = (-1, 3)$, see the top row of **Figure 7.2**.*

Read each matrix as a function and the product is nothing new: with $g(v) = Sv$ and $f(w) = Rw$, the two-step route of matrix multiplication is simply like the composition of functions $f(g(v))$. You first transform v to Sv , afterwards you apply R to Sv – RS simply does both at once. That the brackets do not matter for matrices is also clear here: doing S and then R is one and the same motion however you group it, so $R(Sv) = (RS)v$, written simply RSv . That order matters for matrices – as we check below, $RS \neq SR$ – now stops being strange. This is also the case for functions: if $f(x) = x^2$ and $g(x) = 2x$, then $f(g(x)) = 4x^2 \neq 2x^2 = g(f(x))$. For more on this, see [3Blue1Brown: Matrix multiplication](#).

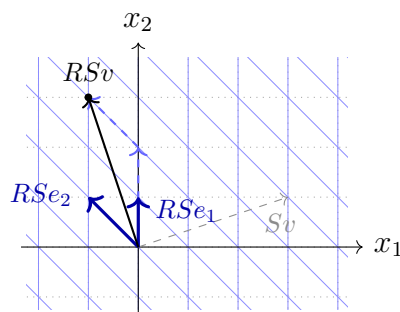
To see that the order really matters, run the same two motions the other way round – turn first, shear afterwards:

$$SR = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}.$$

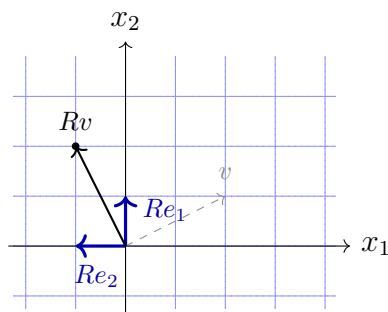
Its columns send e_1 to $(1, 1)$ and e_2 to $(-1, 0)$. Applying both to $v = (2, 1)$ in turn: $Rv = (-1, 2)$, then $S \cdot (-1, 2) = (1, 2)$; and the single matrix SR gets there in one step, since $(SR)v = 2\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1\begin{pmatrix} -1 \\ 0 \end{pmatrix} = (1, 2)$. Comparing the two routes, $(SR)v = (1, 2) \neq (-1, 3) = (RS)v$: the same two motions in the other order leave v somewhere else entirely; see the bottom row of **Figure 7.2**.



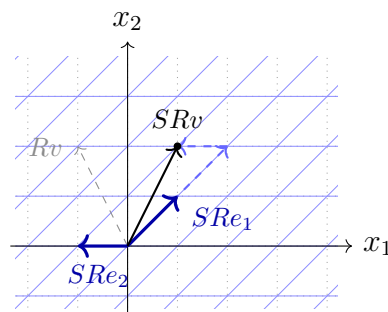
(a) After S : $Sv = (3, 1)$



(b) After R as well: $RSv = (-1, 3)$



(c) After R : $Rv = (-1, 2)$



(d) After S as well: $SRv = (1, 2)$

Figure 7.2: Order matters: shear then turn (top) sends v to $(-1, 3)$, while turn then shear (bottom) sends it to $(1, 2)$

Exercise 7.1. Some exercises on a matrix being a table of landing spots for the basis vectors.

- A transformation sends e_1 to $(2, 0)$ and e_2 to $(0, 3)$. Write down its matrix D , compute Dv for $v = (2, 1)$, and describe in words what D does to the plane.
- A transformation has matrix $\begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix}$. Where does e_2 land? Sketch both landing spots.
- Let R_{-90} be a quarter turn clockwise. Write down its matrix by asking only where e_1 and e_2 land. Explain (in words) why we would expect $R R_{-90}$ to leave every vector where it is? Show that for $v = (2, 1)$ we get $R R_{-90} v = v$.

7.4 Transpose and Inverse

Two concepts that are relevant for solving matrix equations of models, both in economic theory and in econometrics, are those of the *transpose* and the *inverse* of a given matrix.

Example 7.7. (*Transpose*) The transpose of a matrix is the matrix resulting from swapping columns and rows:

$$C := \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad C^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad (7.2)$$

The *inverse* of a matrix A , denoted A^{-1} , is such that $AA^{-1} = I_2$, where I_2 is the 2×2 “identity matrix”: $I_2 := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. For any 2×2 matrix, multiplying it with the identity matrix renders the same matrix – just like multiplying a scalar with the number 1: I_2 is what “doing nothing” looks like as a matrix. You have already met both ideas: in **Exercise 7.1c**) the clockwise quarter turn undoes the anticlockwise one, leaving every vector where it was – so $RR_{-90} = I_2$, that is, R_{-90} “undo” R and $R_{-90} = R^{-1}$.

Example 7.8 (Deriving the Inverse Formula for a 2×2 Matrix). Consider a given 2×2 matrix $M := \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We want to find a matrix $M^{-1} := \begin{bmatrix} x & y \\ z & w \end{bmatrix}$ such that $MM^{-1} = I_2$. We compute the product:

$$MM^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} ax + bz & ay + bw \\ cx + dz & cy + dw \end{bmatrix}.$$

We now require this product to equal the identity matrix:

$$\begin{bmatrix} ax + bz & ay + bw \\ cx + dz & cy + dw \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \iff \begin{cases} ax + bz = 1 \\ ay + bw = 0 \\ cx + dz = 0 \\ cy + dw = 1 \end{cases}$$

Solving this system of equations, we eventually find that the solution is:

$$M^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

provided that $ad - bc \neq 0$. This quantity $ad - bc$ is called the **determinant** of M , denoted $\det(M)$. A square matrix M is **invertible** if, and only if, $\det(M) \neq 0$.

Exercise 7.2. Some basic exercises of matrix multiplication:

- a) Consider the matrix A above. What is $A^\top$? Get $A^\top A$. Is it the same as $AA^\top$?
- b) Apply the inverse formula for A to get A^{-1} . Verify that $AA^{-1} = I_2$.
- c) Consider a vector of residual values from an OLS regression $\varepsilon := (\varepsilon_1, \dots, \varepsilon_n)^\top$. Get $\varepsilon^\top \varepsilon$. Is it the same as $\varepsilon \varepsilon^\top$?

For more on the connection between concepts introduced here and an econometric regression, I thoroughly recommend the [geometric interpretation of OLS](#).

7.5 Systems of Linear Equations

In economics, matrices often describe *linear systems of equations* that arise from equilibrium conditions and/or budget constraints, e.g. from a firm choosing inputs, or a consumer choosing a bundle. Mathematically, we write such a system as

$$Ax = b.$$

For intuition, recall the interpretation of a matrix as a linear transformation from **Example 7.5**: A is a motion of the plane, and b is where that motion sends x . Solving the system means running it backwards — we can see where the vector landed, and we are trying to back out where it started.

ECON APPLICATION: Anna's optimisation: A linear system

Take Anna (**Chapter 6**) with utility $u(x_1, x_2) = \sqrt{x_1 x_2}$, prices $p = (1, 2)$, and income $y = 10$. Later on, we will motivate that her optimal bundle will satisfy two conditions: (i) her budget binds: $x_1 + 2x_2 = 10$, (ii) a *tangency condition* $x_1 = 2x_2$. Together, these form a linear system of two equations in two unknowns:

$$\begin{cases} x_1 - 2x_2 = 0 \\ x_1 + 2x_2 = 10 \end{cases} \iff \underbrace{\begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ 10 \end{bmatrix}}_b \quad (7.3)$$

The inverse exists, so we get one unique solution:

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \frac{1}{4} \underbrace{\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 0 \\ 10 \end{bmatrix}}_b = \begin{bmatrix} 5 \\ 2.5 \end{bmatrix} \quad (7.4)$$

So Anna's optimal bundle is $x^* = (5, 2.5)$, giving utility $u(x^*) = \sqrt{12.5}$.

We often avoid computing A^{-1} directly. In practice, this inverse may not even exist. Yet even without an inverse, a system may still have a solution. More generally, a system of linear equations can have exactly one, multiple or no solution.

Example 7.9 (One solution). *The following system has one (unique) solution:*

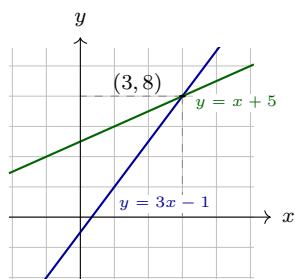
$$\underbrace{\begin{bmatrix} 3 & -1 \\ 1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 1 \\ -5 \end{bmatrix}}_{\mathbf{b}} \iff \begin{cases} 3x - y = 1 \\ x - y = -5 \end{cases} \iff \begin{cases} y = 3x - 1 \\ y = x + 5 \end{cases}$$

Example 7.10 (No solution). *The following system of parallel lines has no solution:*

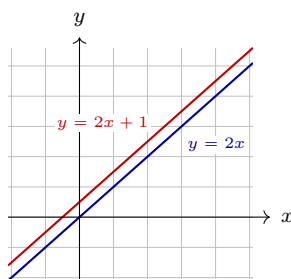
$$\begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \iff \begin{cases} x = 0.5y \\ y - 2x = 1 \end{cases} \iff \begin{cases} y = 2x \\ y = 2x + 1 \end{cases}$$

Example 7.11 (Infinitely many solutions). *We may also have an infinite number of solutions:*

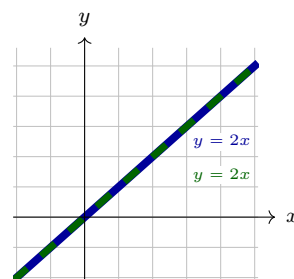
$$\begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \iff \begin{cases} x = 0.5y \\ y - 2x = 0 \end{cases} \iff \begin{cases} y = 2x \\ y = 2x \end{cases}$$



(a) **Example 7.8**



(b) **Example 7.9**



(c) **Example 7.10**

Figure 7.3: Two lines cross once, never, or everywhere

7.6 Gaussian Elimination

For large systems of equations, there is no general shortcut for computing A^{-1} , and solving by variable substitution quickly becomes infeasible beyond small examples. Fortunately, we have a powerful method for solving systems of equations efficiently — even when we don't compute A^{-1} explicitly.

Gaussian Elimination solves the system $A\mathbf{x} = \mathbf{b}$ by applying row operations to the augmented matrix $[A \mid \mathbf{b}]$, step by step, until the left side becomes the identity matrix:

$$[A \mid \mathbf{b}] \longrightarrow [I \mid \mathbf{x}].$$

The final column then gives the solution vector $\mathbf{x}$. In other words, Gaussian elimination produces the same result as computing $\mathbf{x} = A^{-1}\mathbf{b}$ — but without explicitly forming the inverse matrix. It is systematic, scalable, and works for systems with one, none, or infinitely many solutions. A quick Google search will give you the foundations and you can practice it yourself with the exercises below.

In practice, you'll often rely on a computer to do the heavy lifting of solving large systems of equations — but it's important to understand what the computer is doing. Manually applying Gaussian elimination gives you insight into how solutions are constructed step by step, and helps you build intuition for when solutions exist and when they don't.

Exercise 7.3. *Solve the following problems using Gaussian elimination.*

a) *Show that the system below has a unique solution:*

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & -1 \\ 3 & -2 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 9 \end{bmatrix} \iff \begin{cases} x - y + z = 8 \\ 2x + 3y - z = -2 \\ 3x - 2y - 9z = 9 \end{cases} \quad (7.5)$$

b) *Show that the system below has no solution:*

$$\begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} \iff \begin{cases} x - y = 3 \\ 2x - 2y = 7 \end{cases} \quad (7.6)$$

c) *Show that the system below has an infinite number of solutions.*

$$\begin{bmatrix} -1 & -2 & 1 \\ 2 & 3 & 0 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \iff \begin{cases} -x - 2y + z = -1 \\ 2x + 3y = 2 \\ y - 2z = 0 \end{cases} \quad (7.7)$$

8 Metrics

In a vector space V , we can add vectors and scale them by real numbers. But something is missing: there is no way to talk about how *close* two vectors are to each other. To handle distance, we need a different kind of structure: a *metric space*.

8.1 Metric spaces

To formally talk about distance, we need a *metric space*. A metric space does not involve equipping a set with addition or scalar multiplication — just a set of elements X and a *metric* $d(x, y)$: a rule for measuring distances between some $x, y \in X$.

Definition 8.1 (Metric Space). *Let X be a nonempty set. A function $d : X \times X \rightarrow \mathbb{R}$ is a **distance function** or **metric** if for $x, y \in X$, the distance $d(x, y)$ is non-negative, symmetric, equals zero only when the two points are the same, and satisfies the triangle inequality: a detour via a third point is never a short cut. Then (X, d) is a **metric space**.*

You will learn more about what generally constitutes a metric later on, but you already have a basic understanding of the Euclidean distance metric in $\mathbb{R}$ and $\mathbb{R}^2$.

Example 8.1 ($\mathbb{R}$ with the Euclidean Distance). *Define the distance function $d_1(x, y) = |x - y|$. Then $(\mathbb{R}, d_1)$ is a metric space.*

Example 8.2 ($\mathbb{R}^2$ with the Euclidean Distance). *Define the distance function $d_2(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$, where $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$. Then $(\mathbb{R}^2, d_2)$ is a metric space. This is the standard distance in the plane (by the Pythagorean theorem).*

The Euclidean metric is the one you already have intuition for, but it is only *one* rule among many. A metric space is anything satisfying **Definition 8.1**.

We can also define distance functions for more “exotic” sets. Spaces of functions show up throughout economics. In (micro) auction theory, agents choose an optimal bidding function $b^* \in U$. In (macro) dynamic growth models, agents choose optimal consumption-investment policy functions. The properties of these spaces and subsets determine whether an optimal solution can be guaranteed to exist. One metric that can be used on such a space is the *supremum metric*:

Example 8.3. *Consider the set $C[a, b]$ of continuous functions $f : [a, b] \rightarrow \mathbb{R}$ with the **supremum metric** $d_\infty(f, g) = \max\{|f(x) - g(x)| : x \in [a, b]\}$ for each pair of functions $f, g \in C[a, b]$. Then $(C[a, b], d_\infty)$ is a metric space. The metric measures the largest distance between function values over the domain: the supremum is the least upper bound, and a continuous function on $[a, b]$ attains its, so we may write $\max$.*

8.2 Open Balls

You have been taught that $(0, 1)$ is “open” because it does not include its boundary points, and $[0, 1]$ is “closed” because it does. In $\mathbb{R}$, this intuition works well — but what does it even mean for a subset of functions or matrices to be open or closed? To answer these questions, we need a more general and precise notion of these concepts. First, we need a formal definition of what is meant by two elements being “close” to one another.

Definition 8.2 (Open Ball). *Let (X, d) be a metric space, and let $x \in X$ and $r > 0$. The **open ball** around x with radius r is the set*

$$B(x, r) = \{y \in X : d(x, y) < r\}$$

of all points whose distance from x is less than r .

The definition is stated for an arbitrary metric space (X, d) , so it covers every metric at once. What it does *not* do is fix the shape of a ball: that is decided by the metric, not by the definition. The two examples below apply the same definition in two different settings and get two very different pictures.

Example 8.4 (An open ball in $(\mathbb{R}^2, d_2)$). *Under the Euclidean metric d_2 , the ball $B(x', r)$ collects the points lying within distance r of x' , see **Figure 8.1**.*

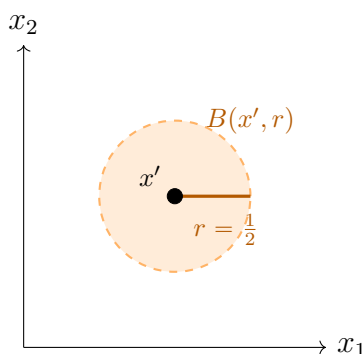


Figure 8.1: An open ball $B(x', r)$ in $\mathbb{R}^2$ around $x' = (1, 1)$ with radius $r = \frac{1}{2}$.

Example 8.5 (An open ball in $(C[a, b], d_\infty)$). *Recall the supremum metric $d_\infty(f, g) = \max\{|f(x) - g(x)| : x \in [a, b]\}$ from **Example 8.3**. Here $B(f, \varepsilon)$ collects every continuous function that stays within ε of f at every point of $[a, b]$. The ball is therefore a tube around the graph of f , drawn in **Figure 8.2**.*

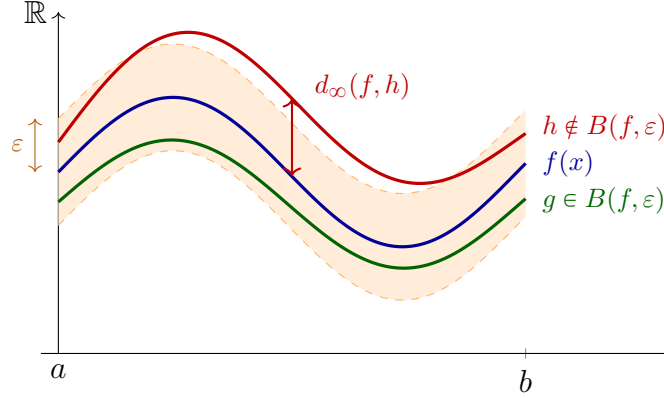


Figure 8.2: An open ball in $(C[a, b], d_\infty)$. The shaded band is $B(f, \varepsilon)$. Function $g(x)$ is within ε of $f(x)$ for any $x \in [a, b]$. Since $d_\infty(f, h) > \varepsilon$, we have $h \notin B(f, \varepsilon)$.

8.3 Openness and closedness

With the notion of open balls in place, we can now define what it means for a point to be *interior* (on the “inside” of the set) and for a set to be *open* or *closed*.

Definition 8.3 (Open and Closed Sets in a Metric Space). *Let (X, d) be a metric space, and let $U \subseteq X$ be a subset in this space.*

- ⊠ A point $u \in U$ is an **interior point** of U if $\exists \varepsilon > 0$ such that $B(u, \varepsilon) \subseteq U$.
- ⊠ A set $U \subseteq X$ is **open** if every point in U is an interior point.
- ⊠ A set $U \subseteq X$ is **closed** if its complement $U^c = X \setminus U$ is open.

Intuitively, for a given subset in a metric space, an *interior point* is an element that one can draw a sufficiently small ball around such that the entire ball is contained in the subset. For a given subset, we say that it is *open* if all elements of the set are interior. If not, then the subset is *not open*. Furthermore, we say that a subset is *closed* if all elements outside of the set are interior to the space outside of the set. If not, then the subset is *not closed*.

Note that sets don’t behave like physical doors: a set can be open, closed, both, or neither. For example, the whole space X and the empty set $\emptyset$ are both open and closed, while the interval $[0, 1) \subset \mathbb{R}$ is neither. Even so, the two notions stay rigidly linked through the complement – U is open exactly when U^c is closed, and vice versa.

Example 8.6 (An interior point). Consider the (blue) subset $(0, \infty) \subset \mathbb{R}$ in **Figure 8.3**. The point $x = 0.1$ is an **interior point** under d_1 , because there exists a small enough (orange) ε -ball for which all elements of the ball are included in $(0, \infty)$, e.g. $\hat{\varepsilon} := 0.05$. But $B(0.1, \varepsilon) \subset (0, \infty)$ need not hold for every $\varepsilon > 0$ for $x = 0.1$ to be interior (e.g. the green ones). It can be shown (see below) that any point in $(0, \infty)$ is interior, so $(0, \infty)$ is open.

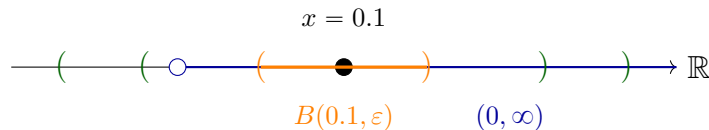


Figure 8.3: The point $x = 0.1$ is an interior point of the interval $(0, \infty)$.

Example 8.7 (A non-interior point). Consider the subset $[0, \infty) \subset \mathbb{R}$. The point $x = 0$ is **not** an interior point under d_1 . For example, in every ball with radius $\varepsilon > 0$, there will be an element $y := -\frac{\varepsilon}{2}$ such that $y \in B(0, \varepsilon)$ but $y \notin [0, \infty)$. Thus, the set is not open.

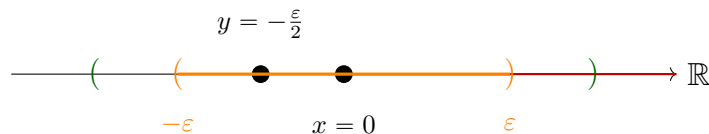


Figure 8.4: The point $x = 0$ is not an interior point of the interval $[0, \infty)$.

A property that can come in handy is also:

Theorem 8.1. Let (X, d) be a metric space. The union $\cup_{i=1}^n A_i$ of any open sets $A_1, \dots, A_n$ is itself an open set. Moreover, it is true for infinitely many sets as well.

Exercise 8.1. Consider some subsets of the real line $\mathbb{R}$, under d_1 .

- Show that $A_1 := (0, \infty)$ is open.
- Show that $A_2 := [0, 1)$ is not open, nor closed.
- Show that $B_1 := [0, 1]$ and $B_2 := [0, \infty)$ are closed.

ECON APPLICATION: Anna's budget set is not open

Anna faces prices $p = (1, 2)$ and has income $y = 10$. Her budget set is

$$A = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + 2x_2 \leq 10\} \subset \mathbb{R}^2$$

The point $x = (3, 2)$ is an interior point under d_2 , as we can draw a small ball around it that remains inside the set ($3 + 4 = 7 < 10$). The point $x' = (2, 4)$, however, lies

ECON APPLICATION (cont.)

on the boundary $x_1 + 2x_2 = 10$, so every ball around it will include points outside the budget set, e.g. $(2 + \frac{\varepsilon}{2}, 4)$. Thus, the budget set is not open.

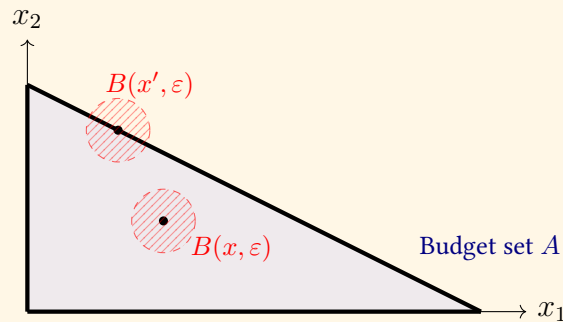


Figure 8.5: Two points in A : $x = (3, 2)$ is interior, $x' = (2, 4)$ is on the boundary.

8.4 Boundary points

We might also define a set's *boundary*. Although there is a trickier formal *definition*, boundary points can be *characterised* as follows: a point u is a boundary point of U if for *every* radius $\varepsilon > 0$, the open ball $B(u, \varepsilon)$ has at least one element in U and at least one element in the complement U^c :

Theorem 8.2. Let (X, d) be a metric space and let $U \subseteq X$ be a subset in this space. A point $u \in X$ is a **boundary point** of U if $\forall \varepsilon > 0$:

$$B(u, \varepsilon) \cap U \neq \emptyset \quad \text{and} \quad B(u, \varepsilon) \cap U^c \neq \emptyset \quad (8.1)$$

Here, note that both notions run the same test – draw a ball around u and ask what it catches – and differ only in the quantifier. Interiority asks whether *some* ball is caught entirely inside U , and because balls nest, $B(u, \varepsilon') \subseteq B(u, \varepsilon)$ for $\varepsilon' < \varepsilon$, one such ball is all we ever need. Being a boundary point asks whether *every* ball straddles U and U^c , so no single ball can ever settle it. The two notions meet in the negation: for a point $u \in U$, being a boundary point is precisely *failing* to be an interior point, since every ball around u already meets U at u itself.

Exercise 8.2. Consider Anna's budget set A . Show that $x' = (2, 4)$ is a boundary point.

Finally, note that a boundary point need not lie in the set. For example, the point 0 is a boundary point of the open set $(0, 1)$, even though $0 \notin (0, 1)$.

8.5 Compactness

Finally, let's introduce two additional properties of a set. First, a set can be *bounded*:

Definition 8.4 (Bounded set). A subset Y of a metric space (X, d) is **bounded** if it is contained in some open ball, i.e. there exist $x \in X$ and $r > 0$ such that $Y \subseteq B(x, r)$.

ECON APPLICATION: Anna's budget set is bounded

Take Anna's budget set A again. For every $x \in A$ we have $0 \leq x_1 \leq 10$ and $0 \leq x_2 \leq 5$, so for example $d_2(x, 0) = \|x\| \leq \sqrt{10^2 + 5^2} = \sqrt{125} < 12$ for every $x \in A$. Hence $A \subseteq B(0, 12)$ and A is bounded, see **Figure 8.6**.

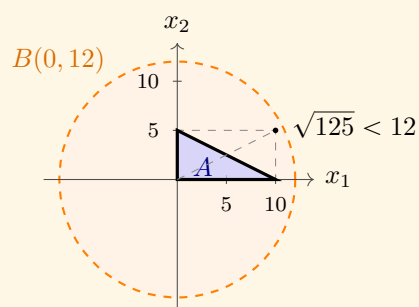


Figure 8.6: Anna's budget set sits inside the open ball $B(0, 12)$, so it is bounded

Second, a property of a set in $\mathbb{R}^n$ that will be crucial later on is *compactness*. A compact set behaves like a finite set, even when it has infinitely many elements. Continuous functions on a finite set are easy – they take finitely many values, so a maximum and a minimum always exist. Compact sets are an infinite version of this, where the same kind of guarantee survives. Later on, you will meet the formal definition of compactness, which works in any metric space and is stated in terms of *open coverings*. For now, we will use a much simpler characterisation that works in $\mathbb{R}^n$:

Theorem 8.3 (Heine-Borel). A subset of $\mathbb{R}^n$, under the Euclidean metric, is **compact** if and only if it is closed and bounded.

So in $\mathbb{R}^n$, compactness is just the conjunction of two properties we understand: a set that contains its own boundary (closed) and does not run off to infinity (bounded) is compact. Later, you will see the most common application of compactness in economics, the *Extreme Value Theorem*: a continuous function attains its maximum on any non-empty compact set. Thus, if an economic agent is maximising a continuous (utility) function over a non-empty compact (budget) set, the problem has a solution.

9 Functions

Consider a *function*, or a *mapping*, $f : X \rightarrow Y$, that takes elements from its *domain* X (a metric space) and maps them into the co-domain Y (also a metric space).

Exercise 9.1. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **strictly increasing** if $x' > x \Rightarrow f(x') > f(x)$. Show that $f(x) = x^2$ is **strictly increasing** on $[0, 1]$, but not on $\mathbb{R}$.

9.1 Images and pre-images

We use the notation f^{-1} with two different meanings, and the cleanest way to separate them is to first ask: *what does a function eat, and what does it spit out?* A function $f : X \rightarrow Y$ eats a *point* of X and returns a *point* of Y . But it also induces two further mappings, which eat *sets* and return *sets*.

Definition 9.1 (Images). Let $f : X \rightarrow Y$. The **image map** $f(\cdot) : \mathcal{P}(X) \rightarrow \mathcal{P}(Y)$ and the **pre-image map** $f^{-1}(\cdot) : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ are defined for $U \subseteq X$ and $V \subseteq Y$, by

$$f(U) = \{y \in Y : y = f(x) \text{ for some } x \in U\}, \quad f^{-1}(V) = \{x \in X : f(x) \in V\}$$

In words: the **image** $f(U)$ collects all function values attained on U , and the **pre-image** $f^{-1}(V)$ collects all points of the domain that are mapped into V .

So the symbols f, f^{-1} do two different jobs, depending on what you feed it:

	eats	returns	always defined?
f , the function	element of X	element of Y	yes
f^{-1} , the inverse	element of Y	element of X	no
$f(\cdot)$, the image	element of $\mathcal{P}(X)$	element of $\mathcal{P}(Y)$	yes
$f^{-1}(\cdot)$, the pre-image	element of $\mathcal{P}(Y)$	element of $\mathcal{P}(X)$	yes

Example 9.1. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^2$. As a function on points it has no inverse, since $f(-2) = f(2) = 4$. As a map on sets both directions always work: $f(\{2\}) = \{4\}$ and $f^{-1}(\{4\}) = \{-2, 2\}$. The round trip is what to notice — $f^{-1}(f(\{2\})) = \{-2, 2\} \neq \{2\}$, so the pre-image mapping is not an inverse of the image mapping, only always defined. Even $f^{-1}(\{-1\}) = \{x \in \mathbb{R} : x^2 = -1\} = \emptyset$ is a legitimate answer, since the empty set is a perfectly good element of $\mathcal{P}(\mathbb{R})$.

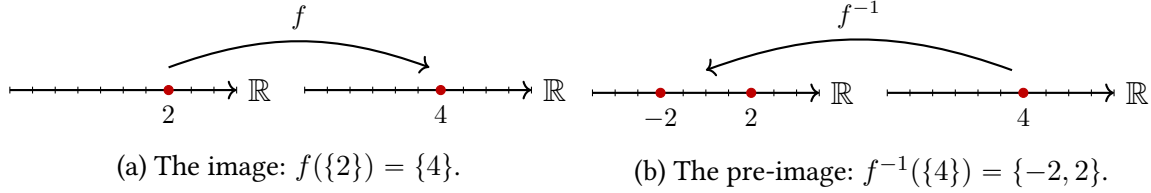


Figure 9.1: Push $\{2\}$ forward and pull it back: we do not return to where we started.

Example 9.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, and consider the subset (in the co-domain) $A = [0, 4) \subseteq \mathbb{R}$. The pre-image of A is all points $x \in \mathbb{R}$ in the domain s.t. $f(x) \in A$:

$$f^{-1}(A) = \{x \in \mathbb{R} : 0 \leq x^2 < 4\} = (-2, 2).$$

The asymmetry of **Example 9.1** survives one level up. Pushing $[0, 2)$ forward gives exactly A , but pulling A back returns $(-2, 2)$, which is strictly larger. **Figure 9.2** draws this on the two copies of $\mathbb{R}$; **Figure 9.3** reads the same pre-image off the graph of f instead.

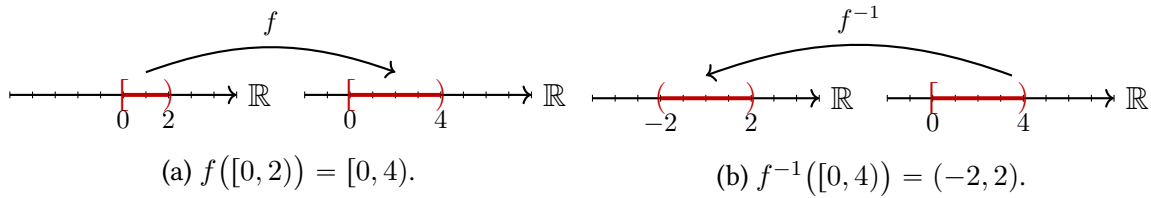


Figure 9.2: The image of $[0, 2)$ and pre-image of $[0, 4)$ under $f(x) = x^2$

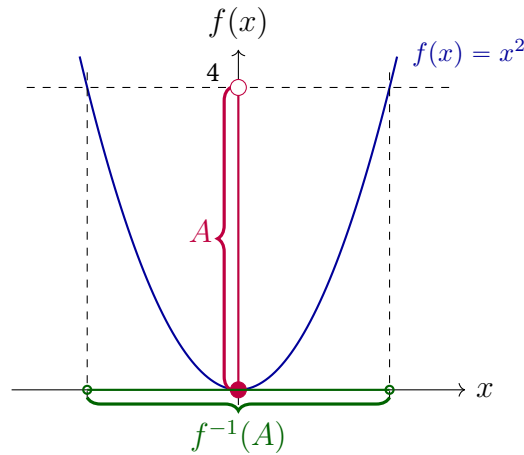


Figure 9.3: The pre-image of **Figure 9.2b** read off the graph of f instead

Exercise 9.2. Now, consider some functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$.

a) Define $f_1(x) := x_1$. Find the image of $[0, 1]^2$ and draw it in $\mathbb{R}$:

$$\{y \in \mathbb{R} : y = f_1(x) \text{ for some } x \in [0, 1]^2\} \subset \mathbb{R} \quad (9.1)$$

b) Find the pre-image of $[0, +\infty)$ under $f_1(x)$ and draw it in $\mathbb{R}^2$:

$$\{x \in \mathbb{R}^2 : f_1(x) \in [0, +\infty)\} \subset \mathbb{R}^2 \quad (9.2)$$

ECON APPLICATION: Anna's indifference curves and upper contour sets

Take Anna with utility $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ such that $u(x_1, x_2) = \sqrt{x_1 x_2}$. Two important pre-images come up everywhere in microeconomics. First, the pre-image of the singleton $\{\bar{u}\}$ under u for some $\bar{u} > 0$ is the set of bundles that give Anna utility exactly $\bar{u}$ — her **indifference curve** at level $\bar{u}$:

$$u^{-1}(\{\bar{u}\}) = \{x \in \mathbb{R}_+^2 : \sqrt{x_1 x_2} = \bar{u}\}.$$

Geometrically, slice Anna's utility surface with three planes. Each cut traces out a curve on the surface, and projecting that curve down to the x_1 - x_2 plane gives the indifference curve at that level, see **Figure 9.4** and **Figure 9.5**.

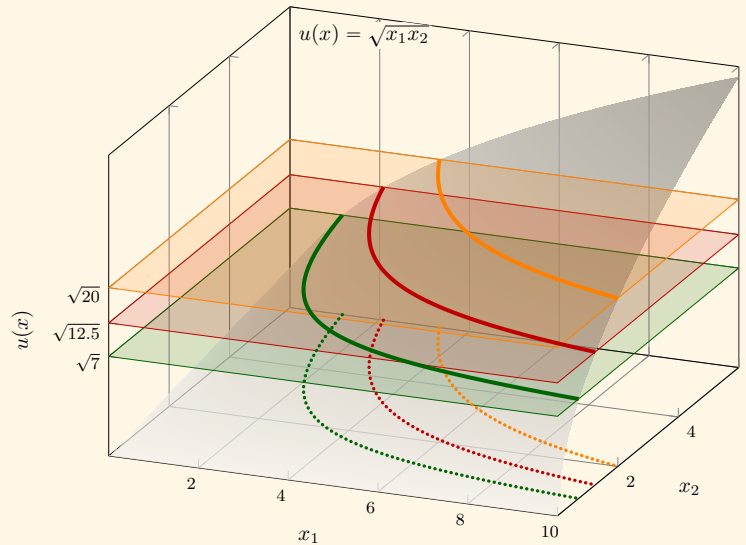


Figure 9.4: Three horizontal slices of $u(x) = \sqrt{x_1 x_2}$, at $\bar{u}_1 = \sqrt{7}$ (green), $\bar{u}_2 = \sqrt{12.5}$ (red) and $\bar{u}_3 = \sqrt{20}$ (orange) and projections down to the x_1 - x_2 plane.

ECON APPLICATION (cont.)

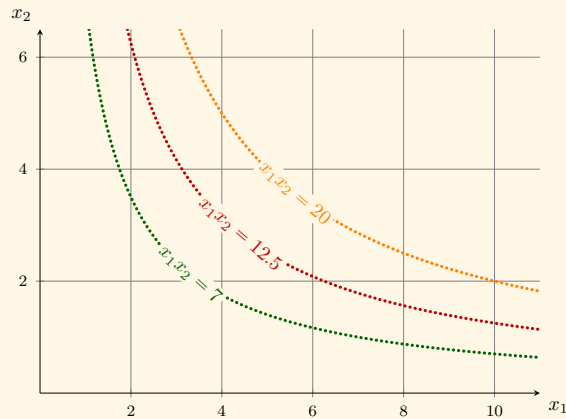


Figure 9.5: The three cuts from **Figure 9.4** projected down in the $x_1 - x_2$ plane

Second, the pre-image of $[\bar{u}, \infty)$ under u is the set of bundles that give Anna utility at least $\bar{u}$ – bundles she weakly prefers to anything on her $\bar{u}$ -indifference curve, see **Figure 9.6**. This is her **upper contour set** at level $\bar{u}$, denoted $U_{\bar{u}}$:

$$U_{\bar{u}} := u^{-1}([\bar{u}, \infty)) = \{x \in \mathbb{R}_+^2 : \sqrt{x_1 x_2} \geq \bar{u}\}.$$

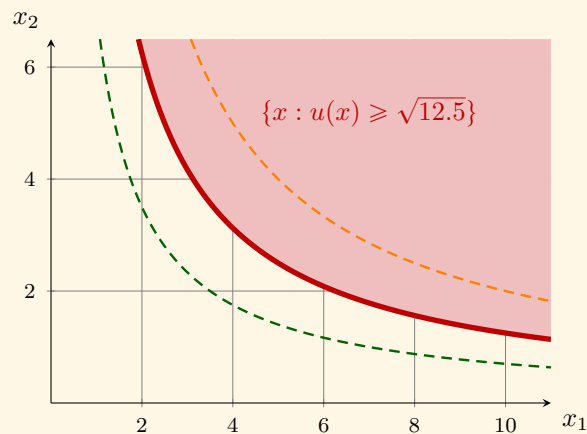


Figure 9.6: The upper contour set $U_{\bar{u}_2}$ at the middle level – every bundle on or above the solid red curve $x_1 x_2 = 12.5$ – and the other two curves of **Figure 9.5**.

9.2 The (ϵ, δ) -definition of continuity

Continuous functions have no sudden jumps in their function values. Look at a point a in the domain with function value $f(a)$. For continuous functions, we have that if $a' \approx a$, then $f(a') \approx f(a)$. Formally, this is captured by the so-called (ϵ, δ) -definition:

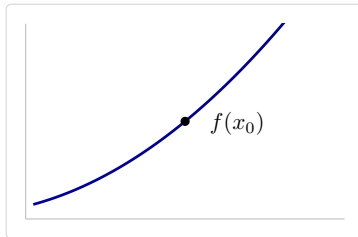
Theorem 9.1 ((ϵ, δ) -definition of continuity). *Let (X, d) and (Y, d') be metric spaces, $f : U \rightarrow Y$ a function on a subset U of X , and let $a \in U$. Then f is **continuous** at a point a if, and only if, for each $\epsilon > 0$ there is a $\delta > 0$ such that*

$$\forall x \in U : d(x, a) < \delta \Rightarrow d'(f(x), f(a)) < \epsilon$$

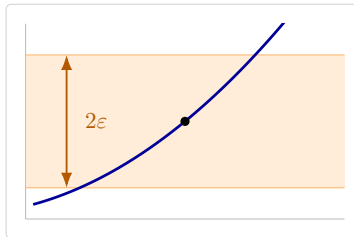
Intuitively, suppose we are standing at the point x_0 on the real line. Someone hands us a desired level of precision $\epsilon > 0$ for the function values. Our task is to construct a precision level $\delta > 0$ for the input such that staying within δ of x_0 guarantees that the function values stay within ϵ of $f(x_0)$. If this is always possible, then f is continuous. See [Figure 9.7](#) below and [3Blue1Brown: Calculus](#) for more intuition.

Exercise 9.3. Show that $f(x) = 2x + 1$ is continuous on $\mathbb{R}$ under d_1 .

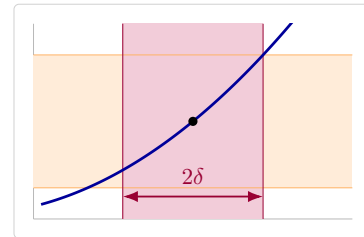
The function $f(x)$ is continuous at x_0



1. We stand at the point x_0 .

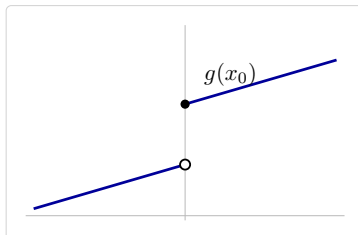


2. You demand this *output* band.

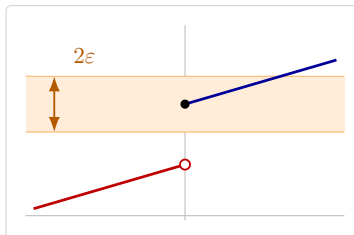


3. I reply: this *input* window.

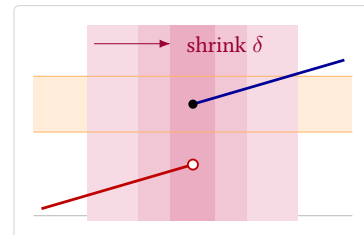
The function $g(x)$ is not continuous at x_0



1. Now the graph jumps at x_0 .



2. A band *narrower* than the jump.



3. No δ is ever small enough.

Figure 9.7: Epsilon-delta definition of continuity, as a “demand” and a “reply”.

As already mentioned, disproving a statement can be relatively easy compared to proving it. When proving that a function is not continuous, it is enough to find a single example of ε where for some x' , we can never find a δ s.t. $d(x', x_0) < \delta \Rightarrow d(f(x'), f(x_0)) < \varepsilon$.

Exercise 9.4. Suppose $x \in \mathbb{R}$ represents a gain/loss on a bet. Consider an agent with a “prospect theory”-inspired utility $u : \mathbb{R} \rightarrow \mathbb{R}$. She feels losses more keenly than wins, and her utility takes a “jump” around zero: she cares about the “winning” and “losing” in itself:

$$u(x) := \begin{cases} \sqrt{x} + 0.5 & \text{if } x \geq 0 \\ -2\sqrt{|x|} & \text{if } x < 0 \end{cases} \quad (9.3)$$

Draw its graph. Then show $u(x)$ is not continuous at 0.

9.3 Continuity and Openness

A useful result is the characterisation of continuous functions in terms of pre-images:

Theorem 9.2 (Continuity and openness/closedness). Let (X, d) and (Y, d') be metric spaces and $f : X \rightarrow Y$. The following three claims are equivalent:

- Function f is continuous;
- Pre-images of open sets are open sets: if $V \subseteq Y$ is open, then $f^{-1}(V)$ is open;
- Pre-images of closed sets are closed sets: if $V \subseteq Y$ is closed, then $f^{-1}(V)$ is closed.

Note that the above theorem can be interpreted as an “if and only if” proposition:

$$f \text{ continuous} \iff V \subseteq Y \text{ open} \Rightarrow f^{-1}(V) \text{ open} \iff V \subseteq Y \text{ closed} \Rightarrow f^{-1}(V) \text{ closed}$$

Note what the theorem does *not* claim: images of open sets need not be open. Continuity says “if inputs are close, then outputs are close”, not the other way around. That is what makes a pre-image of an open set open, but it says nothing about whether f reaches every value near one it attains.

Example 9.3 (Images and openness). Let $f(x) = x^2$. The open set $(4, 9)$ has pre-image $f^{-1}((4, 9)) = (-3, -2) \cup (2, 3)$ – an open set. But image $f((-2, 2)) = [0, 4)$ is not open: 0 is not an interior point, since for every $\varepsilon > 0$ we have $y := -\frac{\varepsilon}{2} \in B(0, \varepsilon)$ but $y \notin [0, 4)$.

Openness of a set: Instead of showing that an open ball can be drawn around every element in the set, one can prove openness using the theorem above.

Exercise 9.5. Take as given that linear functions are continuous, e.g. $f(x_1, x_2) = x_1$. Use a pre-image argument to show that $C := \{x \in \mathbb{R}^2 : x_1 > 0\}$ is open.

Continuity of a function: The epsilon-delta definition of continuity can be quite tricky to work with. At times, a simpler proof is available to show continuity of a function:

Exercise 9.6. Show that the utility function u in **Exercise 9.4** is not continuous by finding an open set V whose pre-image under u is not open.

9.4 Existence of an Optimal Solution

When does an optimisation problem have a solution? Many problems have no maximum. The function $f(x) = x$ on $[0, 1)$ or $[0, \infty)$ has no maximum: any candidate point can be improved upon. EVT gives us sufficient conditions for existence of a maximum.

Theorem 9.3 (Extreme Value Theorem). Let B be a non-empty compact subset of a metric space, and let $f : B \rightarrow \mathbb{R}$ be continuous. Then f attains its maximum on B – i.e. there exist $x^* \in B$ with $f(x^*) \geq f(x)$ for all $x \in B$.

The recipe for showing an economic optimisation problem has a solution is then short: verify (i) the feasible set is non-empty and compact, and (ii) the objective is continuous.

ECON APPLICATION: Anna's optimum exists

Take Anna with any continuous utility $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ over her budget set A , defined in **Chapter 8**.

A is non-empty: Since e.g. $(0, 0) \in A$, we have $A \neq \emptyset$.

A is closed: The complement of A is the union of three sets:

$$A^c = \underbrace{\{x \in \mathbb{R}^2 : x_1 < 0\}}_{A_1^c} \cup \underbrace{\{x \in \mathbb{R}^2 : x_2 < 0\}}_{A_2^c} \cup \underbrace{\{x \in \mathbb{R}^2 : x_1 + 2x_2 > 10\}}_{A_3^c}$$

With a similar reasoning as in **Exercise 9.5**, it can be shown that each A_1^c, A_2^c, A_3^c are open sets. By **Theorem 8.1**, A^c is open, so A is closed.

A is compact: Since A is bounded (shown in **Chapter 8**) and closed in $\mathbb{R}^2$ under d_2 , thus compact, by **Theorem 8.3**.

Conclusion. Since u is a continuous function and A is a non-empty and compact set, u attains a maximum over A , by **Theorem 9.3**.

Example 9.4 (EVT is sufficient, not necessary). Not satisfying the EVT conditions does not mean failing to have a maximum – the theorem is then simply “silent”. Take $f(x) = -(1 - x)^2$ on $(0, \infty)$, a neither closed nor bounded set, thus not compact by **Theorem 8.3**. Yet $f(x) \leq 0$ for every x , with equality only at $x = 1$, so f attains its max at $x^* = 1$.

10 Derivatives

Functions are mappings between metric spaces, and continuity indicates when small changes in the input result in small changes in the output. We now ask a sharper question: *how sensitive* is the output to the input? Suppose I consume x apples. What is the change in utility from consuming an infinitesimally small additional amount of apples? Does such an approximation exist?

10.1 Differentiability

In short, we say that a function is *differentiable* at a certain point if we can locally approximate it with a linear function.

Definition 10.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. We say that f is **differentiable** at a point x if the following limit exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

and we refer to $f'(x)$ as the **derivative** of $f(x)$, also denoted as $\frac{df(x)}{dx}$. If f is differentiable at every point $x \in \mathbb{R}$, we simply say that f is **differentiable**; the derivative is then itself a function $f' : \mathbb{R} \rightarrow \mathbb{R}$.

Exercise 10.1. Using the above definition, show that $f'(x) = 2x$ for $f(x) = x^2$.

We know that $f'(x) = 2x$, so at $x = 1$, the slope is $f'(1) = 2$. So $f(x) = x^2$ can be locally approximated near $x = 1$ by the linear function $g(x) = 2x - 1$, shown in **Figure 10.1**.

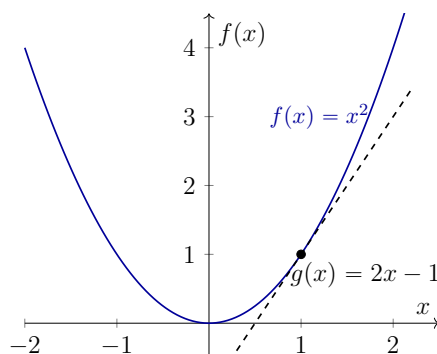


Figure 10.1: The function $f(x) = x^2$ and its linear approximation at $x = 1$.

Differentiability and continuity are related, but only in one direction. If f is differentiable at a point a , then it is also continuous at a : a function that is well approximated

by a straight line near a cannot jump there. The converse fails. Continuity rules out jumps, but it does not rule out *corners*, and at a corner there is no single line to approximate the function by. Differentiability is therefore the strictly stronger property: every differentiable function is continuous, but not every continuous function is differentiable.

Example 10.1. *The function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x) = |x|$ is a continuous function (draw it!) but has no derivative defined at $x = 0$: we can not approximate it by a linear function. We say that $g(x)$ is not differentiable at 0.*

10.2 Partial Derivatives

When a function depends on multiple variables, we can still talk about how it changes — but we examine one variable at a time, holding the others constant. For example, we call the change in function value of some function $f(x, y)$ from an infinitesimally small change in x the *partial derivative* of f with regards to x , denoted $\frac{\partial f(x, y)}{\partial x}$. Note that derivatives of single-variable functions are denoted $\frac{df}{dx}$, while functions of several variables use ∂ to indicate that only one variable (of many) is changing.

Definition 10.2 (Partial Derivative). *Let $f(x_1, \dots, x_n)$ be a real-valued function. The **partial derivative** of f w.r.t. x_i at a point $a := (a_1, \dots, a_n)$ is defined as*

$$\frac{\partial f}{\partial x_i}(a_1, \dots, a_n) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(a_1, \dots, a_n)}{h}.$$

Example 10.2 (Partial Derivatives). *Consider $f(x, y) = -x^2 + 2xy - 2y^2$. To compute the partial derivative with respect to x , we fix y (treating it as a constant) and differentiate:*

$$\begin{aligned} \frac{\partial f}{\partial x} &= \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-(x + h)^2 + 2(x + h)y - 2y^2 - (-x^2 + 2xy - 2y^2)}{h} \\ &= \lim_{h \rightarrow 0} (-2x + 2y - h) = -2x + 2y. \end{aligned}$$

For the partial derivative with respect to y , we get $\frac{\partial f}{\partial y} = 2x - 4y$.

10.3 Gradients

The *gradient* of a function is a vector that collects each partial derivative and thus points in the *direction of steepest ascent*.

Definition 10.3 (Gradient). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a point $a \in \mathbb{R}^n$. The **gradient** of f at a , denoted $\nabla f(a)$, is the vector of all partial derivatives. We write: $\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right)$.

Suppose you are standing at a hill at position (x, y) , where x is east–west coordinate, y is north–south coordinate and $f(x, y)$ is the elevation. The partial derivatives tell you how steep the hill is if you take a tiny step east $\left(\frac{\partial f}{\partial x}\right)$ or north $\left(\frac{\partial f}{\partial y}\right)$. The gradient $\nabla f(x, y)$ points in the direction in which the hill rises most rapidly at (x, y) . The steepest way up is at right angles to the way that keeps you level, so $\nabla f(a)$ is *perpendicular* ($\perp$) to the level set through a .

Example 10.3 (Gradient). For the function $f(x, y) = 2xy - x^2 - 2y^2$, the gradient is $\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (-2x + 2y, 2x - 4y)$. At $(1, 1)$, where $f(1, 1) = -1$, we get $\nabla f(1, 1) = (0, -2)$. Strictly, $\nabla f(1, 1)$ is a vector in the domain $\mathbb{R}^2$ – it says which way to walk in the xy -plane to climb fastest – and drawing it on the graph means walking up the tangent plane at $(1, 1, -1)$, which lifts the horizontal step $(0, -2)$ to the vector $(0, -2, 4)$.

ECON APPLICATION: Anna's gradient

Anna's utility surface is a hill, and $\nabla u(x)$ points the way up it from the bundle x . Reading that arrow against her budget line is what tells us whether she can still do better. Away from x^* , some affordable step raises her utility; at $x^* = (5, 2.5)$ it stands perpendicular to the line, and no affordable step does.

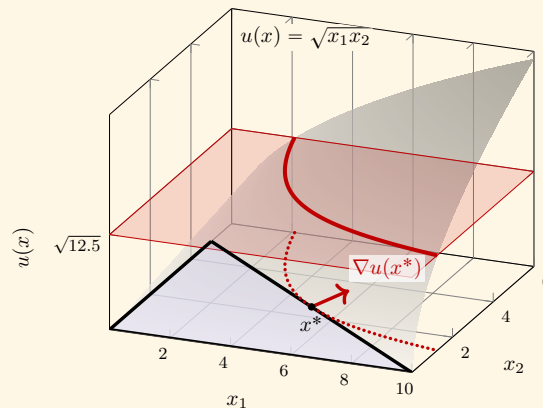


Figure 10.2: **Figure 9.4's** red slice at $\bar{u}^* = \sqrt{12.5}$, now with Anna's budget set drawn on the floor. The dotted red curve is the projected indifference curve, and $\nabla u(x^*)$ sits at $x^* = (5, 2.5)$.

ECON APPLICATION (cont.)

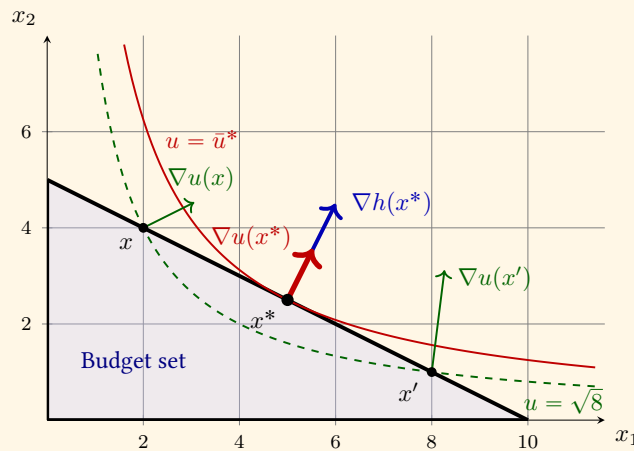


Figure 10.3: The same picture in the x_1 - x_2 plane, with a lower (green) indifference curve at $u = \sqrt{8}$ crossing the budget line at $x = (2, 4)$ and $x' = (8, 1)$, and the constraint gradient $\nabla h(x^*) = (1, 2)$ in blue.

It is worth naming the constraint as a function in its own right, since **Chapter 12** works with it directly. Anna's budget set is $x_1 \geq 0$, $x_2 \geq 0$ and $h(x) \leq 0$, where $h(x) := x_1 + 2x_2 - 10$ is her spending in excess of her income. Being a function, h has a gradient of its own: $\nabla h(x) = (1, 2) = p$, the same at every bundle. So the constraint's steepest-ascent direction is the price vector: it points the way spending rises fastest, at right angles to the budget line and out of the budget set, drawn in blue at x^* in **Figure 10.3**. Reading the two arrows against each other is what settles the problem. On the green curve, $x = (2, 4)$ and $x' = (8, 1)$ give the same utility $\sqrt{2 \cdot 4} = \sqrt{8 \cdot 1} = \sqrt{8}$, so it crosses the budget line twice and Anna can still do better between the crossings. At x^* the two crossings have closed into a tangency and $\nabla u(x^*)$ points along $\nabla h(x^*)$ – the condition we solve in **Chapter 12**.

Exercise 10.2. Consider Anna's utility $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ of the more general form $u(x) = x_1^\alpha x_2^{1-\alpha}$ for some $\alpha \in (0, 1)$.

- Write the gradient $\nabla u(x)$ as a function of $u(x)$. Interpret how the gradient changes with x and α .
- What is the value of the gradient at $x^* = (5, 2.5)$ with $\alpha = 0.5$?

11 Convexity

Convexity is the property of sets and functions that, time and again, makes the difference between economic optimisation problems being well-behaved or hopelessly tangled.

11.1 Convex sets

Intuitively, a set is convex if it has no “dents” or “holes” – pick any two points in the set, and the straight line between them stays inside the set. Formally:

Definition 11.1 (Convex set). *A set $C \subseteq \mathbb{R}^n$ is **convex** if, for any two points $x, y \in C$ and any $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)y \in C$.*

The expression $\lambda x + (1 - \lambda)y$ is a point on the line segment between x and y : at $\lambda = 1$ we are at x , at $\lambda = 0$ we are at y , and as λ slides between the two, we trace out the segment. A rule of thumb to remember (shown below): filled circles are convex, bananas are not!

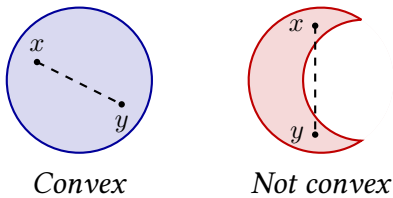


Figure 11.1: A convex set (left) and a non-convex set (right)

ECON APPLICATION: Anna's budget set is convex

Recall Anna's budget set $A = \{x \in \mathbb{R}_+^2 : x_1 + 2x_2 \leq 10\}$. For any two affordable bundles $x, y \in A$ and any $\lambda \in [0, 1]$, the mixed bundle $z = \lambda x + (1 - \lambda)y$ satisfies

$$z_1 + 2z_2 = \lambda(x_1 + 2x_2) + (1 - \lambda)(y_1 + 2y_2) \leq \lambda \cdot 10 + (1 - \lambda) \cdot 10 = 10,$$

and clearly $z_1, z_2 \geq 0$. So $z \in A$, and A is convex.

11.2 Convex and concave functions

Convexity for *functions* is closely related to convexity of sets. For convex functions any chord between two points on the graph lies above the curve. For concave functions, any chord between two points on the curve lies below it.

Definition 11.2 (Convex and concave functions). Let $C \subseteq \mathbb{R}^n$ be a convex set. A function $f : C \rightarrow \mathbb{R}$ is **convex** if, for any $x, y \in C$ and any $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

It is **concave** if $-f$ is convex.

Example 11.1 (Convex and concave functions). The function $f(x) = x^2$ is convex. The function $g(x) = \ln x$ is concave.

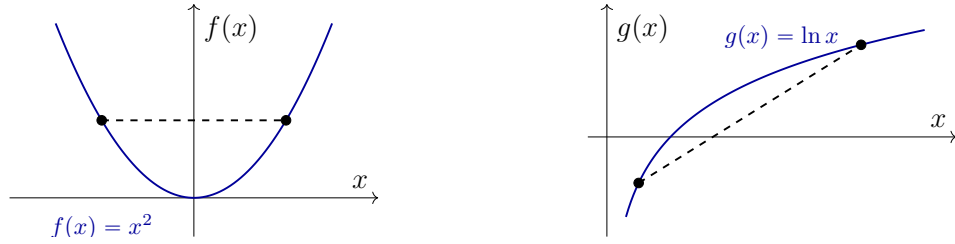


Figure 11.2: Left: $f(x) = x^2$ is convex. Right: $g(x) = \ln x$ is concave.

We formally define the area above $f(x)$ and $g(x)$ as their *epigraph*:

Definition 11.3. The epigraph of a function $f : C \rightarrow \mathbb{R}$ is the set

$$\text{epi}(f) = \{(x, t) \in C \times \mathbb{R} : t \geq f(x)\} \subseteq \mathbb{R}^{n+1}$$

Example 11.2 (Epigraphs). A point of the plane is a pair (x, t) : a first coordinate x , which has to be a legal input, and a second coordinate t , which can be any height. For $f(x) = x^2$ every real number is a legal input, so $C = \mathbb{R}$ and

$$\text{epi}(f) = \{(x, t) \in \mathbb{R}^2 : t \geq x^2\}.$$

For $g(x) = \ln x$, only $x > 0$ are legal inputs, so $\text{epi}(g) = \{(x, t) : x > 0, t \geq \ln x\}$. While $\text{epi}(f)$ is a convex set, $\text{epi}(g)$ is not, see **Figure 11.3**.

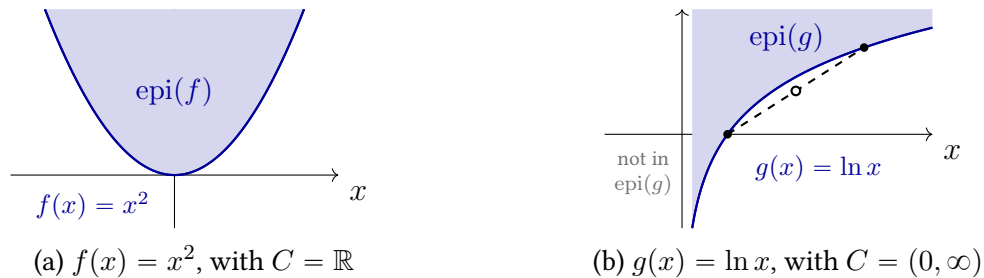


Figure 11.3: The epigraph is the shaded region on and above the graph.

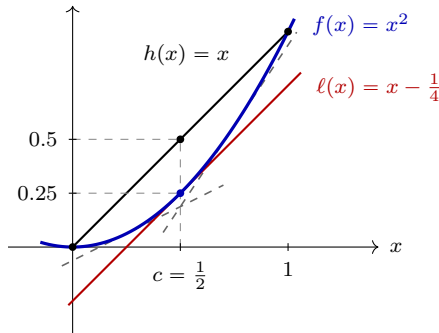
11.3 Three ways to check the shape of a function

Suppose you stumble upon a function $f(x)$ in the wild. How to test if it is convex/concave?

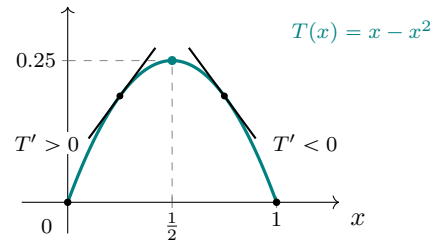
- (i) **Chord definition:** Show that the chord between any two points on the graph lies above (convex) or below (concave) the graph. Often quite tedious!
- (ii) **Epigraph:** A function $f : C \rightarrow \mathbb{R}$ is convex if and only if $\text{epi}(f)$ is a convex set.
- (iii) **Derivative test.** For *differentiable* and *single-variable* functions: f is convex if $f''(x) \geq 0$ everywhere and concave if $f''(x) \leq 0$ everywhere.

The intuition for the derivative test is that f'' measures how the slope changes, and a convex function has a non-decreasing slope: the graph keeps curving upward, so the chord has to sit above it. For intuition, see [3Blue1Brown: Higher order derivatives](#).

Example 11.3 (Why increasing slope puts the chord above the graph). Take $f(x) = x^2$ and the chord between 0 and 1. The chord is $h(x) = x$ and the gap between chord and graph is $T(x) = h(x) - f(x) = x - x^2 = x(1 - x)$. Clearly, $T(x) \geq 0$ for $x \in [0, 1]$, so chord will always be above graph, so f is convex by **Definition 11.2**. But what does that have to do with $f'' \geq 0$? The Mean Value Theorem (introduced in main course) gives us that there exists some $c \in (0, 1)$ such that $f'(c) = h'(c)$. Here, that point is $c = \frac{1}{2}$, so the tangent $\ell(x) = x - \frac{1}{4}$ is parallel to the chord, see **Figure 11.4**. Since f' is increasing, the graph is flatter than the chord left of c and steeper right of it, so the gap grows up to c and shrinks after: $T'(x) = 1 - 2x$ is positive before $\frac{1}{2}$ and negative after. A gap that starts at 0, rises, and falls back to 0 is never negative, so $T \geq 0$ on $[0, 1]$.



(a) the chord, the graph, and the tangent at c



(b) the gap rises, then falls

Figure 11.4: $f(x) = x^2$ with its chord h over $[0, 1]$ and the tangent ℓ at midpoint $c = \frac{1}{2}$. The grey tangents at $\frac{1}{4}$ and $\frac{3}{4}$ are flatter and steeper than the chord, respectively, so the gap $T = h - f$ climbs up to c and falls away after it, never dipping below zero.

Exercise 11.1. Consider Anna’s indifference curve for some utility level $\bar{u}$: this is the set defined as $\{x \in \mathbb{R}_+^2 : \sqrt{x_1 x_2} = \bar{u}\}$. Along this indifference curve, we can describe the choice of x_2 as a function of x_1 (to keep the utility constant) s.t. $x_2 = f(x_1)$. This function is:

$$f(x_1) = \frac{\bar{u}^2}{x_1} \quad (11.1)$$

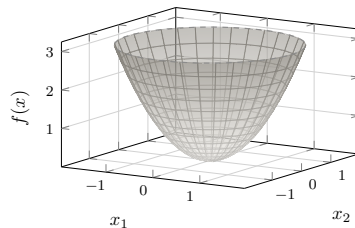
This is a twice-differentiable single-variable function $f : (0, \infty) \rightarrow \mathbb{R}$. Show it is convex with each of the three routes explained above.

Example 11.4 (When the derivative test runs out). The function $g(x) = |x|$ is convex by route (i) — the chord clearly sits above the V-shape. But since g has no second derivative at 0, route (iii) is unavailable.

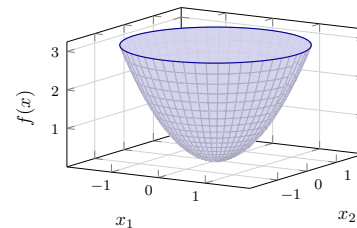
Example 11.5 (A convex function of two variables). We wish to show that $f(x_1, x_2) = x_1^2 + x_2^2$ is convex. The derivative test is unavailable, because f is not single-variable. However, applying some clever algebra, one can show with **Definition 11.2** that for any two points $x, y \in \mathbb{R}^2$ and any weight $\lambda \in [0, 1]$, mixing the inputs costs no more than mixing the outputs:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

This is left as an exercise. After showing that f is a convex function, we have also “for free” shown that $\text{epi}(f)$ — the filled bowl in **Figure 11.5** — is a convex set.



(a) the graph of f



(b) $\text{epi}(f)$

Figure 11.5: The graph of $f(x_1, x_2) = x_1^2 + x_2^2$ is a shell; its epigraph is that shell together with everything on or above it — a bowl, filled

12 Constrained Optimisation

Most of what economic theory does is constrained optimisation: an agent maximising an objective, subject to constraints. While your high-school maths gave you the rule “ $f'(x) = 0$ at optimum”, the situation is trickier with constraints and multivariable functions. The general tool is the *Lagrangian*, formalised through the *Karush–Kuhn–Tucker (KKT) conditions*, covered in great length in the main course.

Here is the intuition of the optimality conditions in the case of Anna in the case of an interior solution. When Anna’s single constraint $h(x) \leq 0$, where $h(x) = x_1 + 2x_2 - 10$, binds at the optimum, the gradient of the objective must be parallel to the gradient of the constraint at the optimum:

$$\underbrace{\nabla u(x^*)}_{\text{direction of fastest utility increase}} = \underbrace{\lambda}_{\text{shadow price}} \underbrace{\nabla h(x^*)}_{\text{direction of cost increase}}, \quad \lambda \geq 0. \quad (12.1)$$

Read economically, the condition drawn in **Figure 12.1** equates two exchange rates. The steepness of Anna’s indifference curve is the rate at which she is *willing* to give up x_2 for a little more x_1 — her *marginal rate of substitution* (MRS), x_2/x_1 here — and the steepness of her budget line is the rate at which the market *lets* her, $p_1/p_2 = \frac{1}{2}$. At $x^* = (5, 2.5)$ the two coincide, so no affordable swap is worth making. Equivalently, the last unit of income buys the same extra utility whichever good it goes to, and that common value is the multiplier λ — the extra utility Anna gets from one more unit of income, i.e. the *shadow price* of her budget. Below, we find x^* with the Lagrangian and a simpler route.

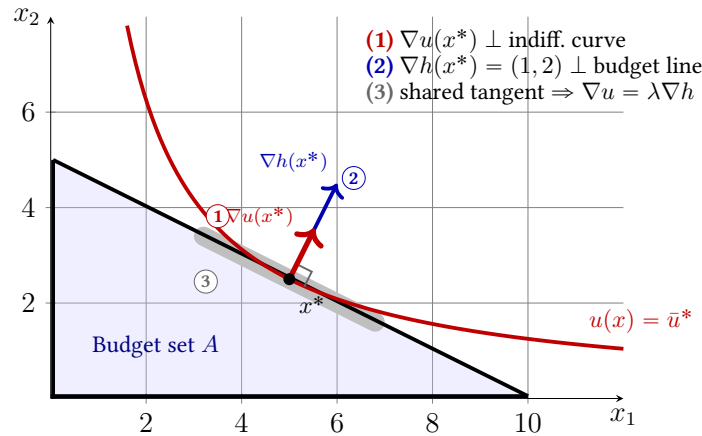


Figure 12.1: At Anna’s optimum x^* , the indifference curve and the budget line share their tangent at x^* . $\nabla u(x^*)$ (red) is perpendicular to the indifference curve and $\nabla h(x^*) = (1, 2)$ (blue) to the budget line, hence parallel: $\nabla u(x^*) = \lambda \nabla h(x^*)$.

This tangency – and that Anna’s MRS equals the price ratio – is not a general law: it holds because Anna’s optimum is *interior*. Corner solutions can break it, and the more general rule of the KKT conditions will be covered later on.

ECON APPLICATION: Anna’s optimal bundle

Anna’s budget set is

$$A := \{x \in \mathbb{R}^2 : x_1, x_2 \geq 0, x_1 + 2x_2 \leq 10\}$$

Anna’s problem is

$$\max_{x \in A} \sqrt{x_1 x_2}.$$

From **Chapter 9**, we know that an optimum exists. Since $u(x_1, 0) = u(0, x_2) = 0$, while there exists a better feasible bundle $u(1, 1) = 1$, the non-negativity constraints will not bind in optimum, which is what licenses the tangency above.

Method 1: Substitution. Her utility is strictly increasing in each good, so the budget constraint will bind in optimum: $x_1 + 2x_2 = 10$. Since the budget binds,

$$\begin{aligned} x_1 &= 10 - 2x_2 \\ \Rightarrow u(x_2) &= \sqrt{(10 - 2x_2)x_2} \end{aligned}$$

Under some conditions, that will be stated more rigorously later on in the course, optimum can be found at:

$$u'(x_2) = \frac{10 - 4x_2}{2\sqrt{(10 - 2x_2)x_2}} = 0 \quad \Rightarrow \quad \begin{cases} x_2 = 2.5, \\ x_1 = 5. \end{cases} \quad (12.2)$$

Method 2: Lagrangian. The Lagrangian is $\mathcal{L} = \sqrt{x_1 x_2} - \lambda(x_1 + 2x_2 - 10)$. The first-order condition $\nabla \mathcal{L} = 0$ translates to:

$$\begin{cases} \frac{1}{2}\sqrt{x_2/x_1} = \lambda \\ \frac{1}{2}\sqrt{x_1/x_2} = 2\lambda \end{cases} \quad \Rightarrow \quad x_1 = 2x_2 \quad (12.3)$$

Substituting back into either FOC gives $\lambda = \frac{1}{2\sqrt{2}} > 0$. A positive multiplier means the budget constraint must be active (this is a *complementary slackness* condition), so $x_1 + 2x_2 = 10$. Combined with $x_1 = 2x_2$, we get $x^* = (5, 2.5)$.

The Lagrangian is the standard tool of constrained optimisation in economics. Substitution is its more elementary cousin, useful when the active constraints are obvious and the problem collapses to a single variable. Later on, you will see solutions with greater rigour and extended to multi-constraint settings via the KKT conditions.

13 Concluding Words

On these pages, we've seen some of the building blocks of mathematics. While it might be difficult to piece everything together, hopefully, you will soon realize that it's all about taking small pieces and putting them together in more and more complex ways. Just like chess games can be very complicated, it all stems from the rules of how to move the different pieces.

Look back, for instance, at Anna. By the time you reached the convexity chapter you had already met every tool her problem requires: bundles as vectors (**Chapter 6**), prices as inner products (**Chapter 6**), her problem as a system of linear equations (**Chapter 7**), the budget set as a closed and bounded subset of $\mathbb{R}^2$ (**Chapter 8**), utility as a continuous function (**Chapter 9**), the gradient as her direction of steepest ascent (**Chapter 10**), and the convexity of her budget set (**Chapter 11**). Mark's first lectures will revisit each piece more rigorously and give you a framework for understanding the environments of economic agents.

I hope you enjoyed this “boot camp” and that you feel more prepared for the Master's program in Economics. I'll see you after the summer, where we will begin our mathematics adventure by reviewing this material together and addressing any remaining questions that may have come up during your work with these notes.

14 Suggested Solutions

Exercise 2.1. The proof goes:

Base case: For $n = 1$, we have $2 = 1(1 + 1) = 2$, so true.

Inductive step: Assume that for $n = k$, we have $2 + 4 + \cdots + 2k = k(k + 1)$.

Induction proof: Using the inductive hypothesis:

$$2 + 4 + \cdots + 2k + 2(k + 1) = k(k + 1) + 2(k + 1) = (k + 1)(k + 2)$$

By induction, the formula holds for all $n \geq 1$.



Exercise 4.1:

($\Rightarrow$) If x is even, then $x = 2k$ for some $k \in \mathbb{Z}$. Squaring both sides,

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2),$$

Since $2k^2 \in \mathbb{Z}$, we have that x^2 is divisible by 2, so x^2 is even.

($\Leftarrow$) If x^2 is even, then $x^2 = 2m$ for some $m \in \mathbb{Z}$. Suppose, for contradiction, that x is odd. Then $x = 2k + 1$ for some $k \in \mathbb{Z}$, so

$$x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

Since $2k^2 + 2k \in \mathbb{Z}$, then x^2 is odd. But this contradicts that x^2 is even.

We conclude that x is even if, and only if, x^2 is even. $\square$

Second part: Suppose, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2} = p/q$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$. By cancelling all common factors, we may choose p and q to have no common factor — so, in particular, *not both even*. We have

$$\sqrt{2} = \frac{p}{q} \quad \Rightarrow \quad 2 = \frac{p^2}{q^2} \quad \Rightarrow \quad 2q^2 = p^2 \quad (14.1)$$

Clearly, p^2 is even (as it is divisible by 2), so p is even. Since p is even, we can write $p = 2m$ for some $m \in \mathbb{Z}$. We have:

$$2q^2 = (2m)^2 \quad \Rightarrow \quad q^2 = 2m^2 \quad (14.2)$$

But then q is also even. This means that both p and q are even. But this is a contradiction, as we assumed that not both p and q were even. We have arrived at a contradiction. $\square$

Exercise 6.1: Proofs for each sub-questions are below.

a): Since $0 \notin W_0$, it does not satisfy all the properties of a subspace. Since a set is a subspace if and only if it satisfies all three properties, we conclude that W_0 is not a subspace of $\mathbb{R}^2$. $\square$

b): Consider $(0, 1) \in W_1$ and $(1, 0) \in W_1$. Their sum $(1, 0) + (0, 1) = (1, 1) \notin W_1$. Since the set is not closed under addition, we conclude by **Theorem 6.1** that it is not a subspace of $\mathbb{R}^2$. $\square$

c): Consider $\alpha' := 0.5 \in \mathbb{R}$ and $x' := (1, 1) \in W_2$. Then $\alpha'x' = (0.5, 0.5) \notin W_2$. Since the subset is not closed under scalar multiplication, we conclude by **Theorem 6.1** that W_2 is not a subspace. $\square$

d): Let us show that the set satisfies all three properties in **Theorem 6.1**.

(i) Since $3 \cdot 0 - 4 \cdot 0 = 0$, we have $(0, 0) \in W_3$.

(ii) For every $x \in W_3$, we have $x_2 = \frac{3}{4}x_1$. Consider some $z, z' \in W_3$. Denote their sum $v := z + z'$. Then

$$v := z + z' = \begin{bmatrix} z_1 + z'_1 \\ z_2 + z'_2 \end{bmatrix} = \begin{bmatrix} z_1 + z'_1 \\ \frac{3}{4}z_1 + \frac{3}{4}z'_1 \end{bmatrix} = \begin{bmatrix} z_1 + z'_1 \\ \frac{3}{4}[z_1 + z'_1] \end{bmatrix} \quad (14.3)$$

Thus $v_2 = \frac{3}{4}v_1$, so $v \in W_3$. We conclude that W_3 is closed under addition.

(iii) For every $x \in W_3$, we have $x_2 = \frac{3}{4}x_1$. Then for $v := \alpha x$, we have $v_1 = \alpha x_1$ and $v_2 = \alpha x_2$. We can write $v_2 = \alpha \frac{3}{4}x_1$. So $v_2 = \frac{3}{4}v_1$. Thus, $v \in W_3$, so the set is closed under scalar multiplication.

We conclude that W_3 is a subspace of $\mathbb{R}^2$. $\square$ **e):** Let us show that the set satisfies all three properties in **Theorem 6.1**.

(i) We can write the zero function $f_0(x) = 0$ as $f_0(x) = 0 + 0 \cdot x$, so $f_0 \in W_4$.

(ii) Consider any two vectors in W_4 and denote them $f(x) = a + bx \in W_4$ and $g(x) = c + dx \in W_4$, where $a, b, c, d \in \mathbb{R}$. Their sum is

$$(f + g)(x) = (a + c) + (b + d)x,$$

where $a + c \in \mathbb{R}$ and $b + d \in \mathbb{R}$. Hence $f + g \in W_4$, so W_4 is closed under addition.

(iii) Let $\alpha \in \mathbb{R}$ and $f(x) = a + bx \in W_4$, where $a, b \in \mathbb{R}$. Then

$$(\alpha f)(x) = \alpha a + \alpha bx,$$

where $\alpha a \in \mathbb{R}$ and $\alpha b \in \mathbb{R}$, so $\alpha f \in W_4$. Thus, W_4 is closed under scalar multiplication.

We conclude that W_4 is a subspace of $P_2(\mathbb{R})$. $\square$

Exercise 7.1 a): The columns of a matrix are the landing spots of e_1 and e_2 , so

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad Dv = 2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

D stretches the plane by a factor 2 horizontally and 3 vertically, leaving every direction pointing where it did.

b): Call the matrix M . Its second column is the landing spot of e_2 , so e_2 lands at $(3, 0)$, while the first column sends e_1 to $(1, -2)$. Note that e_2 lands along the direction e_1 used to point, three units out.

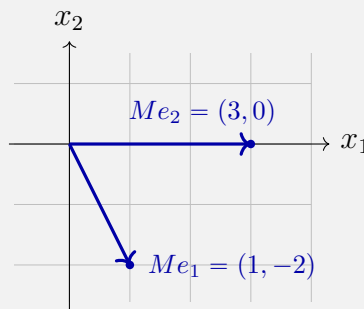


Figure 14.1: Where M sends the two basis vectors

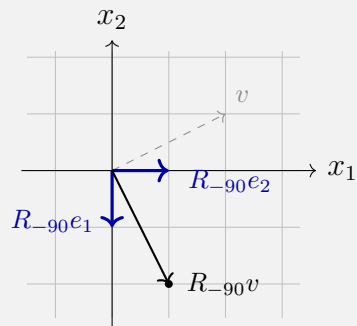
c): A quarter turn clockwise sends $e_1 = (1, 0)$ to $(0, -1)$, and $e_2 = (0, 1)$ to $(1, 0)$. Those two landing spots are the columns:

$$R_{-90} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

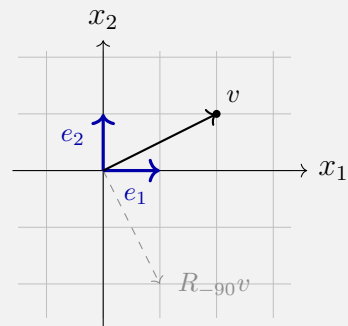
Turning the plane a quarter turn clockwise and then a quarter turn anticlockwise puts every vector back where it started, so together they should do nothing at all — and in a product the matrix nearest the vector acts first, so RR_{-90} is exactly “clockwise, then anticlockwise”. Taking $v = (2, 1)$ one matrix at a time:

$$R_{-90}v = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad R \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = v.$$

The two turns are drawn in **Figure 14.2**. The second panel returns not just v but the basis vectors themselves to where they started.



After R_{-90} : $R_{-90}v = (1, -2)$.



After R as well: $R R_{-90}v = v$.

Figure 14.2: A quarter turn clockwise, then a quarter turn anticlockwise, puts v and the basis vectors back where they started

Exercise 7.2 a) We compute the transpose:

$$A^\top = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \Rightarrow A^\top A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix},$$

while

$$AA^\top = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 11 \\ 11 & 25 \end{bmatrix}.$$

So $A^\top A \neq AA^\top$.

b) Applying the inverse formula for A gives:

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}.$$

We verify that $AA^{-1} = I_2$:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix} = \begin{bmatrix} 1 \cdot (-2) + 2 \cdot 1.5 & 1 \cdot 1 + 2 \cdot (-0.5) \\ 3 \cdot (-2) + 4 \cdot 1.5 & 3 \cdot 1 + 4 \cdot (-0.5) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

c) We have that $\varepsilon^\top \varepsilon \in \mathbb{R}$ is a scalar:

$$\varepsilon^\top \varepsilon = \sum_{i=1}^n \varepsilon_i^2.$$

This product is the *residual sum of squares* — it summarises how far the predicted values are from the observed values. In contrast, $\varepsilon \varepsilon^\top \in \mathbb{R}^{n \times n}$ is an $n \times n$ matrix with elements:

$$(\varepsilon \varepsilon^\top)_{ij} = \varepsilon_i \varepsilon_j.$$

This product is the *outer product* of the residual vector; its expectation $\mathbb{E}[\varepsilon \varepsilon^\top]$ is the *variance-covariance matrix*.

Exercise 7.3 a): Write in matrix form:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 8 \\ 2 & 3 & -1 & -2 \\ 3 & -2 & -9 & 9 \end{array}\right) \quad (14.4)$$

Take $-2R_1$ in second row and $-3R_1$ in third row:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 8 \\ 0 & 5 & -3 & -18 \\ 0 & 1 & -12 & -15 \end{array}\right) \quad (14.5)$$

Take $-5R_3$ in second row:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 8 \\ 0 & 0 & 57 & 57 \\ 0 & 1 & -12 & -15 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 1 & 8 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -12 & -15 \end{array}\right) \quad (14.6)$$

Take $+R_3$ to first row:

$$\left(\begin{array}{ccc|c} 1 & 0 & -11 & -7 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -12 & -15 \end{array}\right) \quad (14.7)$$

Take $+11R_2$ to first row and $+12R_2$ to third row:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -3 \end{array}\right) \quad (14.8)$$

We get:

$$\begin{cases} x = 4 \\ y = -3 \\ z = 1 \end{cases} \quad (14.9)$$

Exercise 7.3 b): Suppose, for contradiction, that the system has a solution. Write in matrix form:

$$\left(\begin{array}{cc|c} 1 & -1 & 3 \\ 2 & -2 & 7 \end{array} \right) \quad (14.10)$$

Now, let's eliminate the first column of the second row by subtracting $2R_1$ from the second row:

$$\left(\begin{array}{cc|c} 1 & -1 & 3 \\ 0 & 0 & 1 \end{array} \right) \quad (14.11)$$

This results in the equation $0 = 1$, which is a contradiction. Therefore, the system of equations has no solution.

c): Write in matrix form:

$$\left(\begin{array}{ccc|c} -1 & -2 & 1 & -1 \\ 2 & 3 & 0 & 2 \\ 0 & 1 & -2 & 0 \end{array} \right) \quad (14.12)$$

To eliminate the first column of the second row, add $2R_1$ to R_2 :

$$\left(\begin{array}{ccc|c} -1 & -2 & 1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & 1 & -2 & 0 \end{array} \right) \quad (14.13)$$

Next, add R_2 to R_3 :

$$\left(\begin{array}{ccc|c} -1 & -2 & 1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (14.14)$$

From the second row, we solve for y in terms of z . Substitute $y = 2z$ into the first equation to express z in terms of x :

$$-x - 2(2z) + z = -1 \quad \Rightarrow \quad x = 1 - 3z \quad (14.15)$$

Thus, the solution for y and z as functions of x is:

$$\begin{cases} y = \frac{2(1-x)}{3} \\ z = \frac{1-x}{3} \end{cases} \quad (14.16)$$

Exercise 8.1: Answers for each subquestion is presented below.

a): Wish to show that each element of A_1 is an interior point: for every $x \in A_1$, there exists some $\varepsilon > 0$ s.t. $B(x, \varepsilon) \subseteq A_1$. Let $x \in (0, \infty)$ and consider the open ball $B(x, \varepsilon) = \{y \in \mathbb{R} : |x - y| < \varepsilon\}$. Then for every point $y \in B(x, \varepsilon)$, we have:

$$|x - y| < \varepsilon \Rightarrow -\varepsilon < x - y < \varepsilon \Rightarrow x - \varepsilon < y < x + \varepsilon.$$

Define $\varepsilon = \frac{x}{2} > 0$. We get $y \in (x - \varepsilon, x + \varepsilon) = (\frac{x}{2}, \frac{3x}{2}) \subset (0, \infty)$. Hence, $B(x, \varepsilon) \subset A_1$. Since each element of A_1 is an interior point, it is open.

b): Consider $0 \in A_2$. For every $\varepsilon > 0$, we have $-\frac{\varepsilon}{2} \in B(0, \varepsilon)$ but $-\frac{\varepsilon}{2} \notin A_2$, so $B(0, \varepsilon) \not\subseteq A_2$. So 0 is not an interior point. Thus, A_2 is not open.

Suppose A_2 was closed. Then A_2^c is open. But 1 is not an interior point to A_2^c , as $1 - \frac{\varepsilon}{2} \in B(1, \varepsilon)$ but $1 - \frac{\varepsilon}{2} \notin A_2^c$. We conclude that A_2 is not closed.

c): $B_1 = [0, 1]$ is closed since its complement is open, as it is the union of two open intervals $B_1^c = (-\infty, 0) \cup (1, \infty)$. Similarly, B_2 is closed, since its complement $B_2^c = (-\infty, 0)$ is open.

Exercise 8.2: Every ball around $x' = (2, 4)$ will include points outside the budget set, e.g. $(2 + \frac{\varepsilon}{2}, 4) \in B(x', \varepsilon) \cap A^c$. In addition, $x' \in B(x', \varepsilon) \cap A$. Since $B(x', \varepsilon) \cap A^c \neq \emptyset$ and $B(x', \varepsilon) \cap A \neq \emptyset$ for every $\varepsilon > 0$, we conclude by **Theorem 8.2** that x' is a boundary point.

Exercise 9.1: Consider points $z := 0$ and $z' := -1$. We have $z > z'$. However, we have $f(z') > f(z)$, since $f(-1) = 1$ and $f(0) = 0$. Thus, f is not strictly increasing.

Next, suppose f was not strictly increasing on $[0, 1]$. That means that there exist some pair $x', x \in [0, 1]$ such that $x' = x + k$ for some $k > 0$ but

$$(x + k)^2 \leq x^2 \quad \Rightarrow \quad 2xk + k^2 \leq 0 \quad (14.17)$$

which is a contradiction, since $2xk + k^2 > 0$ when $x \in [0, 1]$.

Exercise 9.2 a): If $x_1 \in [0, 1]$ and $x_2 \in [0, 1]$, it follows that $f_1(x) = x_1 \in [0, 1]$, so the image is contained in $[0, 1]$. Conversely, every $y \in [0, 1]$ is attained, since $(y, 0) \in [0, 1]^2$ and $f_1(y, 0) = y$. Hence the image is exactly $[0, 1]$.



Figure 14.3: Image of $[0, 1]^2$ under $f_1(x) = x_1$.

b): The pre-image is the “half-space” $\{x \in \mathbb{R}^2 : x_1 \geq 0\}$.

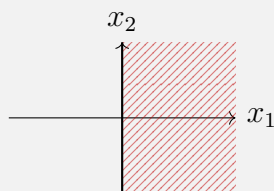


Figure 14.4: Pre-image of $[0, +\infty)$ under $f_1(x) = x_1$.

Exercise 9.4: Graph is shown below.

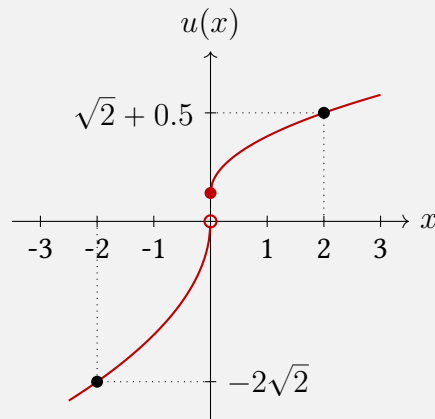


Figure 14.5: Prospect Theory Utility

If $u()$ is continuous at 0, then for each $\varepsilon > 0$, there is a $\delta > 0$ such that each $x \in X$ with $d(x, 0) < \delta$ has $d(u(x), u(0)) < \varepsilon$. Since this must hold for every $\varepsilon > 0$ if $u()$ is continuous, we simply need to find *one* counter-example. Consider $\varepsilon := \frac{1}{2}$ and let us pick the point $x' := -\frac{\delta}{2}$, so $d(x', 0) < \delta$. Then $d(u(x'), u(0)) = d(-\sqrt{2\delta}, \frac{1}{2}) = \sqrt{2\delta} + \frac{1}{2} > \varepsilon$. We conclude that $u()$ is not continuous.

Exercise 9.3: Fix a point $x_0 \in \mathbb{R}$. For a given $\varepsilon > 0$, we wish to find a characterisation $\delta(\varepsilon)$, i.e. a δ as a function of ε , s.t. for every point $x \in \mathbb{R}$, we have

$$d(x, x_0) < \delta \quad \Rightarrow \quad d(f(x), f(x_0)) < \varepsilon. \quad (14.18)$$

First, note that our δ may not be contingent on the point x . Second, note that we wish to prove the implication (14.18): *if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$* . So we are only concerned with what happens when $|x - x_0| < \delta$. For the case $|x - x_0| < \delta$, written under d_1 , we wish to find a δ s.t.

$$|x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon. \quad (14.19)$$

Note that $|f(x) - f(x_0)| = |(2x + 1) - (2x_0 + 1)| = 2|x - x_0|$. It follows that $|f(x) - f(x_0)| < 2\delta$. To ensure that $|f(x) - f(x_0)| < \varepsilon$, we set

$$\delta := \frac{\varepsilon}{2}.$$

Under this choice of δ , the implication (14.18) follows. Since our choice of x_0 was arbitrary, this holds for every point on $\mathbb{R}$. This finalises the proof.

Exercise 9.5: The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ s.t. $f(x_1, x_2) := x_1$ is linear, thus continuous.

Argument I: Note that we can write $C = \{x \in \mathbb{R}^2 : f(x) \in (0, \infty)\}$, where $f(x) = x_1$. C is the pre-image of the open set $(0, \infty)$ under a continuous function f , thus open.

Argument II: The complement C^c is the pre-image of a closed set $(-\infty, 0]$ under a continuous function f , thus closed. Since the complement of C is closed, we conclude that C is open.

Exercise 9.6: Consider the open set $V := (0, 2.5)$. The pre-image of V under u is $u^{-1}(V) = \{x \in \mathbb{R} : u(x) \in (0, 2.5)\} = [0, 4)$, which is not an open set as the point 0 is not interior. See **Figure 14.6** for a visual interpretation of this argument.

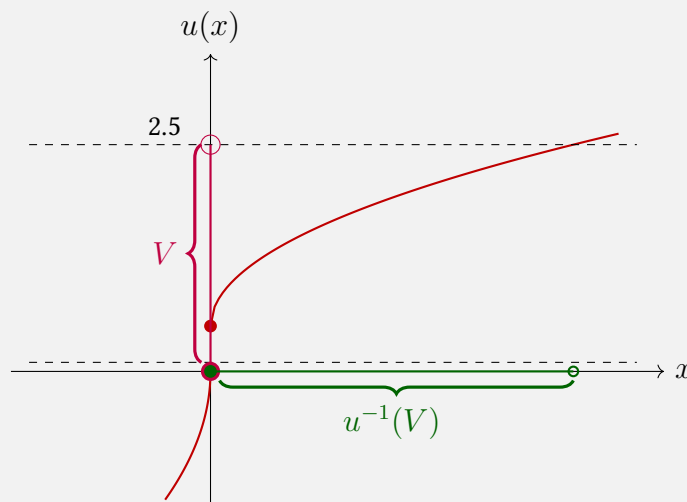


Figure 14.6: Pre-image $u^{-1}(V) = [0, 4)$ of open set $V = (0, 2.5)$ is not open.

Exercise 10.1: Using the definition of a derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x \quad (14.20)$$

Exercise 10.2:

a): The partial derivatives are:

$$\frac{\partial u}{\partial x_1} = \alpha \cdot \frac{u(x_1, x_2)}{x_1} \quad \text{and} \quad \frac{\partial u}{\partial x_2} = (1 - \alpha) \cdot \frac{u(x_1, x_2)}{x_2}.$$

So the gradient at $(x'_1, x'_2) > 0$ is:

$$\nabla u(x'_1, x'_2) = \left(\alpha \cdot \frac{u(x'_1, x'_2)}{x'_1}, (1 - \alpha) \cdot \frac{u(x'_1, x'_2)}{x'_2} \right).$$

Writing the first component as $\alpha \cdot u(x'_1, x'_2)/x'_1$, the coefficient on u/x'_1 rises with α : holding the utility level fixed, a higher α shifts marginal utility towards good 1, reflecting that α is the output elasticity of x_1 . For a given level of x'_2 , the x_1 component decreases in x'_1 , due to diminishing marginal returns.

b): At $\alpha = 0.5$, the utility level is $u(5, 2.5) = \sqrt{12.5}$. The gradient takes the value

$$\nabla u(5, 2.5) = \left(0.5 \cdot \frac{\sqrt{12.5}}{5}, 0.5 \cdot \frac{\sqrt{12.5}}{2.5} \right) = \frac{\sqrt{2}}{4} (1, 2) \quad (14.21)$$

Exercise 11.1: We verify convexity via each route in turn.

Chord definition: Using **Definition 11.2**, we wish to show that for any $x, x' \in (0, \infty)$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)x') \leq \lambda f(x) + (1 - \lambda)f(x') \quad (14.22)$$

We start from something we know to be true and work *towards* the claim. Clearly,

$$\lambda(1 - \lambda)(x - x')^2 \geq 0 \quad (14.23)$$

Adding xx' to both sides and factoring (check the algebra yourself!):

$$xx' \leq xx' + \lambda(1 - \lambda)(x - x')^2 = (\lambda x' + (1 - \lambda)x)(\lambda x + (1 - \lambda)x') \quad (14.24)$$

Dividing both sides by the strictly positive number $xx'(\lambda x + (1 - \lambda)x')$:

$$\frac{1}{\lambda x + (1 - \lambda)x'} \leq \frac{\lambda x' + (1 - \lambda)x}{xx'} = \frac{\lambda}{x} + \frac{1 - \lambda}{x'} \quad (14.25)$$

Finally, multiplying both sides by $\bar{u}^2 > 0$:

$$\frac{\bar{u}^2}{\lambda x + (1 - \lambda)x'} \leq \lambda \frac{\bar{u}^2}{x} + (1 - \lambda) \frac{\bar{u}^2}{x'} \quad (14.26)$$

which is exactly $f(\lambda x + (1 - \lambda)x') \leq \lambda f(x) + (1 - \lambda)f(x')$: the chord lies above the graph. $\square$

Derivative test: The function has a positive second derivative $f''(x) = 2\bar{u}^2/x^3 > 0$, thus convex.

Exercise 11.1 (cont.). Epigraph: The epigraph of $f(x) = \frac{\bar{u}^2}{x}$ is

$$E := \left\{ (x, t) \in (0, \infty) \times \mathbb{R} : t \geq \frac{\bar{u}^2}{x} \right\} \quad (14.27)$$

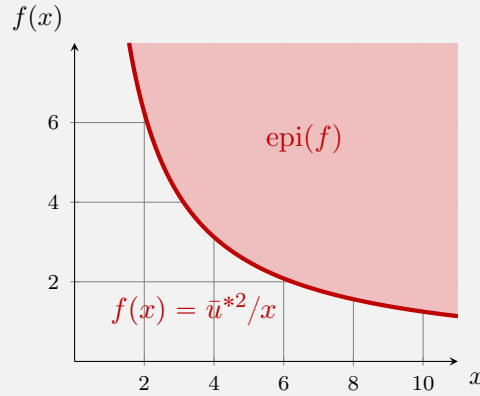


Figure 14.7: The epigraph of $f(x) = \bar{u}^2/x$ on $\mathbb{R}_+$ (shaded red), drawn at Anna's own level $\bar{u}^* = \sqrt{12.5}$: the set of all points lying weakly above the graph of f – her upper contour set of **Figure 9.6**.

Using **Definition 11.3**, we show that E is a convex set. Take any two points $(x, t), (x', t') \in E$ and any $\lambda \in [0, 1]$. By definition of E , we have $t \geq \frac{\bar{u}^2}{x}$ and $t' \geq \frac{\bar{u}^2}{x'}$. The mixed point is $\lambda(x, t) + (1 - \lambda)(x', t') = (\lambda x + (1 - \lambda)x', \lambda t + (1 - \lambda)t')$, and its second coordinate satisfies

$$\lambda t + (1 - \lambda)t' \geq \lambda \frac{\bar{u}^2}{x} + (1 - \lambda) \frac{\bar{u}^2}{x'} \quad (14.28)$$

$$\geq \frac{\bar{u}^2}{\lambda x + (1 - \lambda)x'} \quad (14.29)$$

where the first line uses $t \geq \frac{\bar{u}^2}{x}$ and $t' \geq \frac{\bar{u}^2}{x'}$, and the second is exactly the chord inequality established above. So the mixed point lies in E , and E is convex. Since the epigraph is convex, f is convex.

Finally, note that the epigraph coincides with an upper-contour set in the (x_1, x_2) domain. Later in the course, you will learn the connection between the convexity of upper contour sets and a property of the utility function called *quasiconcavity*.

A The chord and f''

Why should $f'' \geq 0$ force the chord above the graph? Start with a case we can check by hand.

Example A.1 (The gap between chord and graph). For $f(x) = x^2$ the chord over $[0, 1]$ is the line through $(0, 0)$ and $(1, 1)$, that is $h(x) = x$. Measure the gap between chord and graph:

$$T(x) = h(x) - f(x) = x - x^2 = x(1 - x) \geq 0 \quad \text{on } [0, 1],$$

zero at both ends and widest at $x = \frac{1}{2}$, see **Figure A.1**.

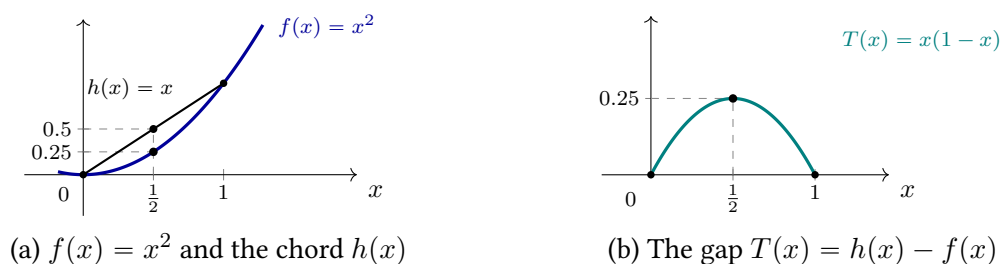


Figure A.1: The chord h over $[0, 1]$ and the gap $T = h - f$ for $f(x) = x^2$

That worked only because $x - x^2$ factorises, which is special to $f(x) = x^2$ and says nothing about f'' . The general case needs one theorem.

Theorem A.1 (Mean value theorem). If f is continuous on $[a, b]$ and differentiable on (a, b) , there is a $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Somewhere strictly inside, the graph's own slope equals the chord's; see **Figure A.2**.

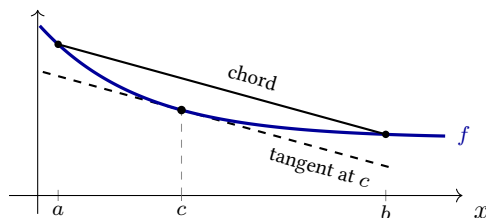


Figure A.2: the tangent at c is parallel to the chord

Proposition A.1 (The derivative test). *Let $C \subseteq \mathbb{R}$ be a convex set and let $f : C \rightarrow \mathbb{R}$ be twice differentiable with $f''(t) \geq 0$ for every $t \in C$. Then f is convex in the sense of **Definition 11.2**.*

Proof. We must show that the inequality of **Definition 11.2** holds for every pair of points and every weight. So take any pair $a, b \in C$ with $b > a$ and any $\lambda \in [0, 1]$, and write $z = \lambda a + (1 - \lambda)b$ for a point on the line segment between them. We wish to show that

$$\underbrace{f(z)}_{\text{the function}} \leq \underbrace{\lambda f(a) + (1 - \lambda)f(b)}_{\text{the chord}}. \quad (\text{A.1})$$

Step 1: Define gap function: Let h be the chord over $[a, b]$ between $f(a)$ and $f(b)$, the straight line with $h(a) = f(a)$ and $h(b) = f(b)$. At point z , we have $h(z) = \lambda f(a) + (1 - \lambda)f(b)$. We denote the gap between f and h as $T(z) = h(z) - f(z)$, so condition (A.1) is equivalent to $T(z) \geq 0$.

Step 2: Gap function has non-decreasing slope: Consider any two points $s, t \in (a, b)$ such that $t > s$. **Theorem A.1** applied to T' gives a $c \in (a, b)$ between such that:

$$T'(t) - T'(s) = T''(c)(t - s) \quad (\text{A.2})$$

Since h is a straight line, $h'' = 0$, so $T''(c) = -f''(c)$ and thus $T'(t) - T'(s) \leq 0$. Thus, by $f'' > 0$, we have that the slope of T never increases.

Step 3: Contradiction: Now, suppose, for contradiction, that the chord h fell below f at some point, so $T(z_0) < 0$ for some $z_0 \in (a, b)$. Since $T(a) = T(b) = 0$, it must mean that the gap fell from 0 to something negative, then climbed back to 0. **Theorem A.1** turns each of those two trips into an actual slope somewhere inside it:

$$\begin{aligned} T'(\alpha) &= \frac{T(z_0) - T(a)}{z_0 - a} = \frac{T(z_0)}{z_0 - a} < 0 && \text{for some } \alpha \in (a, z_0), \\ T'(\beta) &= \frac{T(b) - T(z_0)}{b - z_0} = \frac{-T(z_0)}{b - z_0} > 0 && \text{for some } \beta \in (z_0, b). \end{aligned}$$

So the slope of the gap is bigger at the later β than at the earlier α , contradicting Step 2. Hence $T \geq 0$ on $[a, b]$, so f is convex. $\square$